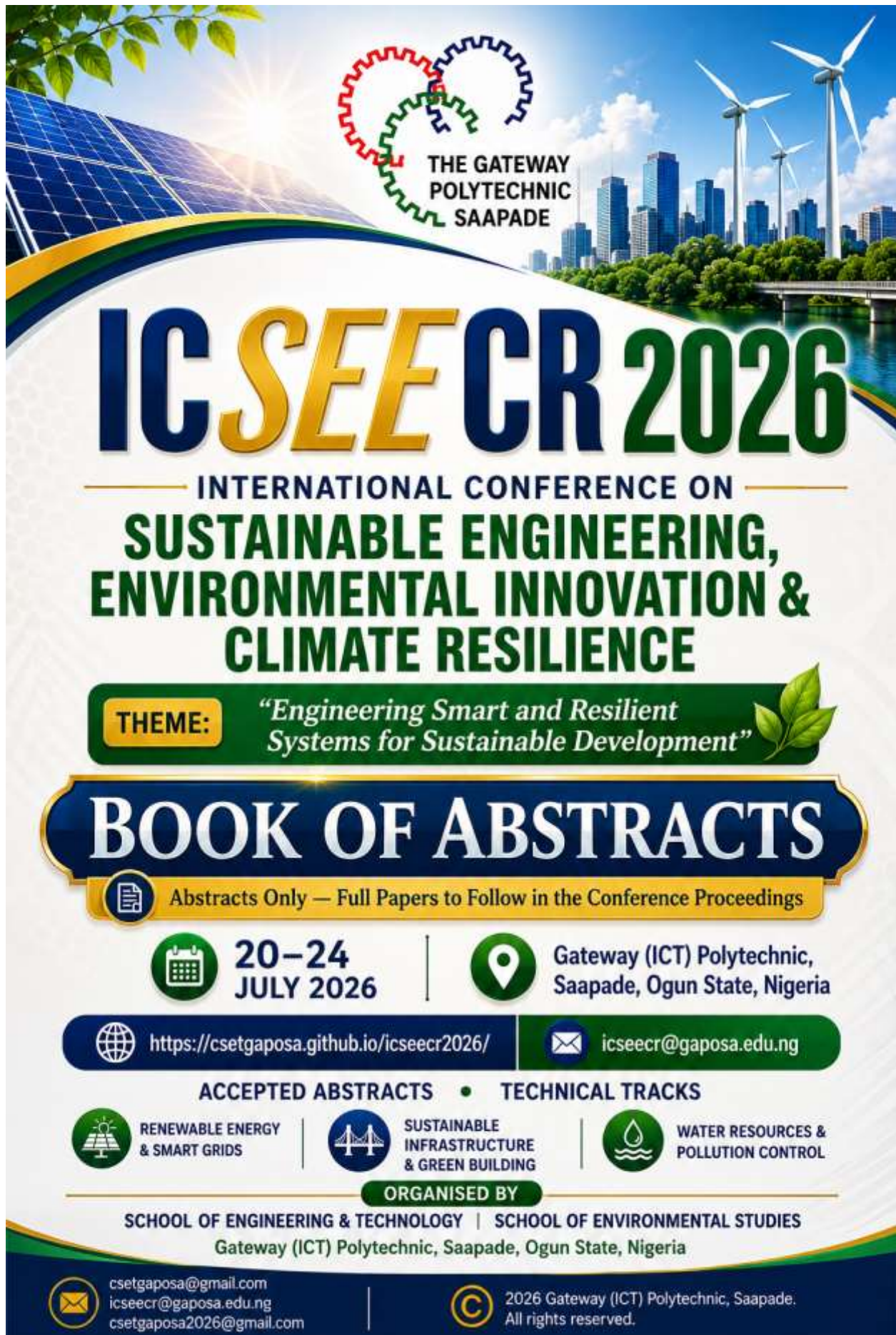




The background of the cover features a collage of sustainable technology and nature: solar panels on the left, a city skyline in the center, and wind turbines on the right, all under a bright sun with rays.

**THE GATEWAY
POLYTECHNIC
SAAPADE**

ICSEECR 2026

— INTERNATIONAL CONFERENCE ON —
**SUSTAINABLE ENGINEERING,
 ENVIRONMENTAL INNOVATION &
 CLIMATE RESILIENCE**

THEME: *“Engineering Smart and Resilient
 Systems for Sustainable Development”*

BOOK OF ABSTRACTS

 Abstracts Only — Full Papers to Follow in the Conference Proceedings

 **20–24
JULY 2026** |  Gateway (ICT) Polytechnic,
Saapade, Ogun State, Nigeria

 <https://csetgaposa.github.io/icseecr2026/> |  icseecr@gaposa.edu.ng

ACCEPTED ABSTRACTS • TECHNICAL TRACKS

 RENEWABLE ENERGY & SMART GRIDS	 SUSTAINABLE INFRASTRUCTURE & GREEN BUILDING	 WATER RESOURCES & POLLUTION CONTROL
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ORGANISED BY

SCHOOL OF ENGINEERING & TECHNOLOGY | SCHOOL OF ENVIRONMENTAL STUDIES
 Gateway (ICT) Polytechnic, Saapade, Ogun State, Nigeria

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ICSEECR 2026

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Accepted Abstracts | Technical Tracks

- *Renewable Energy & Smart Grids (T-01)* | • *Sustainable Infrastructure & Green Building (T-03)* | • *Water Resources & Pollution Control (T-06)*

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Editorial Foreword

On behalf of the Technical Programme Committee and the Local Organising Committee of the International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience (ICSEECR 2026), it is our honour to present this Book of Abstracts for the conference scheduled for 20–24 July 2026, at Gateway (ICT) Polytechnic, Saapade, Ogun State, Nigeria.

This volume assembles the peer-reviewed abstracts of all presentations accepted for the conference. Every abstract was subjected to a rigorous double-blind review process by members of the Technical Programme Committee, evaluated against criteria of originality, methodological soundness, relevance to the conference theme, and alignment with the United Nations Sustainable Development Goals (UN SDGs).

The contributions span three technical tracks — Renewable Energy & Smart Grids, Sustainable Infrastructure & Green Building, and Water Resources & Pollution Control — reflecting the breadth of interdisciplinary research required to address the global climate and sustainability crisis. Authors represent academic institutions, research centres, and industry organisations across Nigeria and beyond.

Abstracts are arranged by technical track and, within each track, in the order accepted for presentation. Each entry is identified by a unique submission ID (ICSEECR-2026-XXX). Duplicate and resubmitted entries have been consolidated so that only the most recent accepted version of each paper appears in this volume. Where an author did not supply keywords with their submission, a concise set of index terms has been derived by the editors from the author's own abstract text and is marked accordingly.

Full conference proceedings, comprising complete reviewed papers, will be published on the official conference website (<https://csetgaposa.github.io/icseecr2026/>) following the post-conference review and editorial cycle.

We extend our sincere appreciation to all contributing authors for their scholarly rigour, to the reviewers for their diligence and expertise, and to the administration of Gateway (ICT) Polytechnic Saapade for their institutional support and commitment to the advancement of sustainable engineering education and research in Nigeria.

The Technical Programme Committee
ICSEECR 2026 — 20–24 July 2026

Note: The views and findings expressed in the abstracts reproduced in this volume are those of the respective author(s) and do not necessarily reflect the position of Gateway (ICT) Polytechnic Saapade or the organising institutions of the conference.

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CONFERENCE OFFICIALS

ICSEECR 2026 — International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience



Dr. Sanni Kehinde Oseni

Chief Host — Rector, Gateway (ICT) Polytechnic, Saapade



Engr. Dr. Kadiri Kamoru Oluwatoyin, FNSE

*Chief Guest of Honour & Academic Lead Paper Presenter
— Rector, The Federal Polytechnic, Offa*



Engr. Felix Adewale Odeleye

*Industrial Lead Paper Presenter — Nuclear Power
Construction Engineer, United Kingdom*



**Esv.(Dr). Samson Efuwape Agbato
(FNIVS, RSV, MRICS)**

Special Guest

KEYNOTE SPEAKERS

Distinguished expertise from academia and industry across three continents



Prof. Emeritus Alireza Baghai-Wadji
Keynote Speaker — University of Cape Town, South Africa



Prof. Y. L. Shuaib-Babata
Keynote Speaker — University of Ilorin, Nigeria



Dr. Rilwan Kayode Apalowo
Keynote Speaker — Universiti Sains Malaysia



Dr. Arc. Oluseyi Israel Abatan
Keynote Speaker — Moshood Abiola Polytechnic, Nigeria



Dr. Emmanuel S. Babatunde
Keynote Speaker — Achievers University, Owo, Nigeria



Chhuon Kong, Ph.D.
Keynote Speaker — Institute of Technology of Cambodia (ITC), Cambodia



Dr. Adeyemi Olusola
Keynote Speaker — York University, Canada



Dr. Adedoyin Mary A.
*Keynote Speaker — Lagos State University,
Nigeria*

KEYNOTE ABSTRACTS

Talk titles and abstracts below are placeholders pending submission by each keynote speaker.

Prof. Emeritus Alireza Baghai-Wadji

Keynote Speaker — University of Cape Town, South Africa

Date & Time: Day 2 — Tue, 21 Jul 2026, 10:50–11:20 WAT (Keynote Address 1) — Main Auditorium

Talk Title: *Self-regularized Method of Moments: Taming Infinities & Zooming into Near Fields*

Abstract: The conventional wisdom in computational engineering (COMENG) suggests that zooming into near fields is tantamount to tempering singularities by taming infinities. The question of whether infinities of various cardinalities are real in nature is as profound as the quest for deciphering the internal structure, twist, spin, or chirality of elemental particles in quantum fields. Thereby, scrutiny of models of idealized forces, e.g., the point charge, and revisiting concepts in mathematical physics, e.g., generalized functions, integrability, and summability, are not merely curious academic pursuits. This lecture is a guided tour for COMENG enthusiasts who wish to learn about the intricate design of problem-specific multi-physics modeling and simulation techniques. It is demonstrated that linear partial differential equations (PDEs) in mathematical physics are diagonalizable. Diagonalized PDEs in the spectral domain are eigenvalue equations. The application of this principle, in conjunction with the theory of generalized functions, leads to the construction of problem-tailored Dirac delta-like functions, yielding the associated self-regularized dyadic Green's functions. Two regularization techniques, algebraic and exponential, are introduced. The calculation of mutual interaction elements in the Method of Moments (MoM) is reduced to the calculation of self-action elements, which is shown to amount to merely sampling smooth functions that can be conveniently precomputed. The method is referred to as the SR-MoM (Self-regularized MoM). The presentation also includes the design of customized basis functions such as Green's function-induced wavelets and frames, Toeplitz B-Spline Wannier functions, the non-standard finite-difference method, and the Discrete Taylor Transform and Inverse Transform. The style of the presentation is axiomatic, rigorous, and based on simple examples, with an inductive storytelling approach intended to guarantee clarity of thought and accessibility.

Engr. Prof. Yusuf Lanre Shuaib-Babata

Keynote Speaker — University of Ilorin, Nigeria

Date & Time: Day 2 — Tue, 21 Jul 2026, 11:20–11:50 WAT (Keynote Address 2) — Main Auditorium

Talk Title: *Engineering Smart and Resilient Systems for Sustainable Development: Challenges, Opportunities and Future Perspectives in Nigeria*

Abstract: In Nigeria, many engineering challenges that threaten sustainable national development include insecurity, inadequate infrastructure, unreliable energy supply, industrial underdevelopment, rapid urbanization, poor waste management, flooding, environmental degradation, climate change impacts, increasing pressure on natural resources, and digital infrastructure deficits. The growth in population, wide gaps in technology, limited investment in research and innovation, and weak maintenance culture further compounded these challenges. Proffering solutions to these complex issues requires the development and deployment of smart, adaptive, resilient, and sustainable engineering systems capable of withstanding environmental, economic, and technological shocks while ensuring long-term national growth and improved quality of life. The ways to address the Nigeria's developmental challenges through emerging technologies (such as Internet of Things (IoT), Artificial Intelligence (AI), Big Data Analytics, Digital Twin technology, Geographic Information Systems (GIS), renewable energy systems, smart manufacturing (Industry 4.0), Building Information Modelling (BIM), and resilient infrastructure engineering are examined in this keynote paper. Emphasizes were made on the strategic role of Nigerian engineers in designing innovative, locally adaptable, and cost-effective engineering solutions that promote energy security, climate resilience, industrial competitiveness, sustainable transportation, smart cities, food security, water resource management, and environmental sustainability. Critical barriers to achieving resilient engineering systems in Nigeria were further identified. Emerging opportunities arising from the African Continental Free Trade Area (AfCFTA), digital transformation, green economy initiatives, renewable energy expansion, local content development, youth innovation ecosystems, and Nigeria's growing startup and technology sectors were revealed. To position engineering as a critical enabler of resilient infrastructure, economic growth, technological advancement, and sustainable national development in Nigeria, some strategic future directions were proposed.

Dr. Rilwan Kayode Apalowo

Keynote Speaker — Universiti Sains Malaysia

Date & Time: Day 3 — Wed, 22 Jul 2026, 10:30–11:00 WAT (Keynote Address 3) — Main Auditorium

Talk Title: *Computational Engineering for Sustainable Engineering Systems: From Multiphysics Simulation to Climate-Resilient Design*

Abstract: Computational engineering has become a vital tool for designing reliable, energy-efficient, and climate-resilient engineering systems in an era shaped by rapid technological innovation and mounting environmental challenges. This keynote presents how advances in multiphysics simulation, computational modelling, and design optimization are shaping the next generation of sustainable engineering systems. Drawing on a decade of research spanning computational mechanics, structural integrity, electronic packaging reliability, and thermal management, the presentation demonstrates how validated computational models have been applied to predict structural damage, characterize multiphysics failure mechanisms, improve the reliability of electronic packaging, and optimize the thermal performance of semiconductor systems. The keynote further illustrates how integrating numerical modelling with experimental validation enables engineers to reduce development time, enhance product reliability, extend service life, improve energy efficiency, and minimize environmental impact. The presentation also discusses emerging opportunities in AI-assisted engineering, physics-informed machine learning, digital twins, and smart manufacturing, highlighting the expanding role of computational engineering in advancing climate-resilient design and sustainable engineering systems.

Dr. Arc. Oluseyi Israel Abatan

Keynote Speaker — Moshood Abiola Polytechnic, Nigeria

Date & Time: Day 2 — Tue, 21 Jul 2026, 12:15–12:45 WAT (Keynote Address 4) — Main Auditorium

Talk Title: *Africa's Demographic Transition: Advancing Sustainability, Innovation And Climate Resilience In The Built Environment Through Multidisciplinary And Interdisciplinary Collaboration Being A Keynote Address Delivered At The Joint Conference Organised By The School Of Environmental Studies And The School Of Engineering Of The Gateway (Ict) Polytechnic, Saapade, Ogun State*

Abstract:

Population dynamics have always shaped civilizations. Throughout history, societies have risen when they successfully anticipated demographic shifts and invested in appropriate infrastructures, institutions and innovation. Conversely, when societies permit population growth to outpace planning, governance and resource management, they groan and struggle. Africa is home to the fastest and youngest growing population in the world. By 2050, one out of every 4 persons in the world would be an African. These changes are neither accidental nor temporary but a long-term demographic transition that is expected to shapen virtually every aspect of lives. A rapidly growing population can be disastrous if not accompanied by quality education, productive employment, good governance, technological innovation and a sustainable built environment. Presently, in most parts of Africa, population growth is reflected in overstretched hospitals, overcrowded classrooms, congested highways, expanding informal settlements, increasing housing demands, flood disasters and a growing need for resilient infrastructure. These realities are too complex for one profession, institution or a sector to tackle. Therefore, a sustainable future for Africa should not depend on merely what each discipline knows but on how effectively those disciplines work collaboratively. This is the essence of a multidisciplinary and interdisciplinary collaboration of this nature.

Dr. Emmanuel S. Babatunde

Keynote Speaker — Achievers University, Owo, Nigeria

Date & Time: Day 3 — Wed, 22 Jul 2026, 11:00–11:30 WAT (Keynote Address 5) — Main Auditorium

Talk Title: [to be confirmed]

Abstract: [to be supplied by the speaker]

Chhuon Kong, Ph.D.

Keynote Speaker — Institute of Technology of Cambodia (ITC), Cambodia

Date & Time: [to be scheduled]

Talk Title: [to be confirmed]

Abstract: [to be supplied by the speaker]

Dr. Adeyemi Olusola

Keynote Speaker — York University, Canada

Date & Time: [to be scheduled]

Talk Title: Building Green, Building Smart, AI and Our Climate Future

Abstract: Our planet is undergoing rapid, far-reaching change, with profound implications for the ecosystems that sustain us. In response, sustainability, clean energy, and green infrastructure have rightly become central pillars of climate action. Increasingly, artificial intelligence is being positioned alongside them as a powerful new tool for building smarter, greener systems. But this promise deserves scrutiny, not blind adoption. As we integrate AI into our climate response, we must resist the temptation to treat it as an unqualified solution. AI itself carries a significant environmental footprint: its growing use is driving up demand for water and energy, adding pressure on the very planet it is meant to help protect. If left unexamined, this tension risks undermining the sustainability gains we seek. This talk makes the case for a more deliberate path forward, one where building green and building smart are pursued together, with clear eyes on AI's costs as well as its capabilities. This balance is especially critical for emerging economies like Nigeria, where infrastructure decisions made today will shape both climate resilience and technological trajectory for decades to come. The choices we make now must serve both a sustainable and a smart future, not one at the expense of the other.

Dr. Adedoyin Mary A.

Keynote Speaker — Lagos State University, Nigeria

Date & Time: [to be scheduled]

Talk Title: [to be confirmed]

Abstract: [to be supplied by the speaker]

NATIONAL ANTHEM

Nigeria we hail thee,
Our own dear native land,
Though tribe and tongue may differ,
In brotherhood, we stand,
Nigerians all, and proud to serve
Our sovereign Motherland.

Our flag shall be a symbol
That truth and justice reign,
In peace or battle honour'd,

And this we count as gain,
To hand on to our children
A banner without stain.

O God of all creation,
Grant this our one request,
Help us to build a nation
Where no man is oppressed,
And so with peace and plenty
Nigeria may be blessed.

NATIONAL PLEDGE

I pledge to Nigeria, my country
To be faithful, loyal and honest
To serve Nigeria with all my strength
To defend her unity
And uphold her honour and glory
So help me God.

OGUN STATE ANTHEM

Ise yaaa, ise yaaa,
Omo Ogun, Ise yaaa,
Olodumare Ise ma ya o ee
Ise yaaa,
Eyin omo Ogun ese giri
Ipo fun wa lati see
Omo rere kii sa seee
Omo ogun ki s'olee
Ebere gb'eru oo
Olorun nbe fun wa
E tera m'ose oo ise ya
Omo Ogun ise yaa

GAPOSA ANTHEM

Great GAPOSA! (2ce)
The pride of Ogun State
Citadel for qualitative learning
Conducive environment
Free from violence
We raise great achievers
Gateway Poly Saapade
ICT Standard
We are the best Polytechnic
Great GAPOSA!
I love GAPOSA!!
Raising a standard for this great nation
Great! Great!! Great!!!
GAPOSA!

RECTOR'S WELCOME SPEECH

Goodwill message from the Rector, Gateway (ICT) Polytechnic, Saapade, for ICSEECR 2026

It is my great honour and privilege to warmly welcome you to the International Conference on Sustainable Engineering, Environmental Innovation and Climate Resilience (ICSEECR 2026). I am deeply delighted to join you in this remarkable gathering of distinguished scholars, researchers, industry professionals, policymakers, and development partners committed to advancing sustainable solutions for a resilient future.

I wish to express my profound appreciation to the Management of the Polytechnic, every committee, department, and teams whose unwavering commitment, professionalism, and collective efforts continue to advance the vision and mission of our institution. Your dedication, collaboration, and steadfast support have been instrumental in making our Polytechnic's pursuit of academic excellence and sustainable development.

The Urgency of the Conference Theme

This year's conference, themed "Engineering Smart and Resilient Systems for Sustainable Development," comes at a time when nations across the world are confronted with unprecedented environmental, technological and socio-economic challenges.

The trajectory of sustainable development requires us to design transdisciplinary, tech-driven solutions to achieve environmental, economic, and social longevity. Issues like climate change, environmental degradation, rapid urbanisation, energy insecurity and resource depletion continue to demand innovative, collaborative and sustainable solutions. The responsibility to respond to these challenges rests significantly on engineers, scientists, researchers, policymakers, innovators and educational institutions.

Engineering has always been a catalyst for human advancement. Today, however, the expectation goes beyond developing infrastructure and technologies; we are called upon to create systems that are intelligent, adaptable, environmentally responsible and resilient enough to withstand present and future challenges. Sustainable development can only be achieved when innovation is combined with environmental stewardship, sound policy implementation and inclusive partnerships.

This conference therefore provides an invaluable platform for scholars and professionals to exchange ideas, present cutting-edge research findings, explore emerging technologies and foster collaborations capable of addressing global environmental and engineering challenges.

Our Commitment to a Sustainable Future

As an institution committed to academic excellence, technological advancement and national development, we recognise the importance of research-driven innovation in shaping sustainable societies. Just recently, we observed World Environment Day under the theme "Inspired by Nature," an event where our students showcased innovations using environmental waste to promote sustainable development. We remain dedicated to promoting interdisciplinary research, strengthening industry-academia partnerships and producing graduates equipped with the skills required to solve sustainable real-world problems through innovation and entrepreneurship.



Dr. Sanni Kehinde Oseni
Rector

A Charge to the Delegates

I encourage every participant to take full advantage of each session, keynote presentation, discussion and networking opportunity that this conference offers. Let us engage in meaningful dialogue, challenge existing assumptions, share practical experiences and develop actionable recommendations that will contribute to sustainable engineering practices and climate resilience.

Moving Forward

Our expectation is that the deliberations during this conference will generate innovative solutions capable of improving infrastructure development, renewable energy systems, environmental management, smart cities, circular economy initiatives, climate adaptation strategies and sustainable industrial development across Africa and beyond.

I wish to specially appreciate our keynote speakers, invited resource persons, sponsors, development partners, members of the organising committee and everyone whose commitment and hard work have made this conference a reality. Your dedication to advancing knowledge and promoting sustainable development is highly commendable.

In conclusion, I sincerely commend the Deans of the Schools of Engineering and Environmental Studies, and members of the Planning Committee for their exceptional dedication, hard work, and commitment to the successful organisation of this conference. May the deliberations at this conference inspire innovative solutions and lasting collaborations for sustainable development.

As we commence this important gathering, I urge us all to remain open-minded, collaborative and solution-oriented. Together, we can engineer smarter systems, protect our environment, strengthen climate resilience and build sustainable communities for generations yet unborn.

It is therefore my honour and privilege to formally declare the 2026 International Conference on Sustainable Engineering, Environmental Innovation and Climate Resilience (ICSEECR 2026) open.

Thank you for your attention, and I wish you all a productive, insightful and rewarding conference.

Dr. Sanni Kehinde Oseni

Rector, Gateway (ICT) Polytechnic, Saapade

WELCOME ADDRESS — SCHOOL OF ENGINEERING & TECHNOLOGY*Welcome message from the Dean, School of Engineering & Technology*

It is my singular honour and pleasure to welcome you — distinguished scholars, industry leaders and delegates travelling from across the globe — to the International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience (ICSEECR 2026), proudly co-hosted by the School of Engineering & Technology, Gateway (ICT) Polytechnic, Saapade.

Sustainable engineering stands today as the defining discipline of our century. As the world confronts the accelerating realities of climate change, resource scarcity and rapid urbanisation, it falls to engineers to conceive the smarter, cleaner and more resilient systems the future demands — from renewable energy and intelligent grids to next-generation infrastructure and sustainable manufacturing. This is not merely a technical challenge; it is a generational responsibility, and one this conference meets with rigour and ambition.

It is with immense pride that this School convenes such an exceptional community of researchers, students and industry professionals, whose work in the pages that follow exemplifies world-class scholarship and bold innovation in equal measure. I warmly encourage every delegate to engage fully with the sessions ahead — to interrogate ideas, forge lasting collaborations, and carry these insights forward into practice that changes lives and shapes a more sustainable world.

My sincere gratitude goes to our esteemed Rector, our Chief Guest of Honour, our distinguished Keynote Speakers and every author whose scholarship has made this gathering possible. I wish you all a truly superb, inspiring and world-class conference.



Engr. Oyebola B. O.
Dean

Engr. Oyebola B. O.

Dean, School of Engineering & Technology, Gateway (ICT) Polytechnic, Saapade

WELCOME ADDRESS — SCHOOL OF ENVIRONMENTAL STUDIES*Welcome message from the Dean, School of Environmental Studies*

On behalf of the School of Environmental Studies, it is my honour to welcome you to the International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience (ICSEECR 2026).



Mr. Lawal A. W.
Dean

The environment we share is under mounting pressure, from land degradation and pollution to the accelerating effects of climate change. Addressing these challenges demands the kind of interdisciplinary collaboration this conference was built to foster, bringing together environmental scientists, planners, engineers and policymakers around a shared commitment to sustainability.

The research gathered in this Book of Abstracts, spanning sustainable infrastructure, green building, water resources and pollution control, reflects the depth of scholarship our contributors bring to these questions. I am confident the days ahead will spark new ideas and partnerships that extend well beyond this conference.

I extend my sincere appreciation to the Organising Committee, our Rector, our esteemed guests and every participant joining us, and I wish you all a rewarding and memorable conference.

Mr. Lawal A. W.

Dean, School of Environmental Studies, Gateway (ICT) Polytechnic, Saapade

CONFERENCE AT A GLANCE

Prince Dapo Abiodun Leadership Auditorium, Gateway (ICT) Polytechnic, Saapade

DAY	MORNING	AFTERNOON
Mon 20 Jul	Arrival & Registration	Arrival & Registration
Tue 21 Jul	Opening Ceremony, Chief Guest & Keynote Addresses 1–4	Keynote Address 4, Industrial Lead Paper, Group Photograph
Wed 22 Jul	Keynote Addresses 5–7	Parallel Technical Sessions (T-01, T-03, T-06) & Poster Session
Thu 23 Jul	Parallel Technical Sessions continue	Closing Ceremony & Awards
Fri 24 Jul	Departure	Departure

A detailed session-by-session programme follows on the next page.

CONFERENCE PROGRAMME

Prince Dapo Abiodun Leadership Auditorium, Gateway (ICT) Polytechnic, Saapade • 20–24 July 2026

Full details of the paper presentation schedule for each session are available on the conference website:
<https://csetgaposa.github.io/icseecr2026/#program>

DAY 1 — MONDAY, 20 JULY: ARRIVAL & REGISTRATION

Arrival, check-in and registration desk open at the hostel / secretariat. No paper presentations scheduled.

DAY 2 — TUESDAY, 21 JULY: OPENING CEREMONY & KEYNOTE SESSION

TIME	SESSION
08:00	Arrival & Registration <i>Secretariat Desk</i>
10:30	Welcome & Opening Remarks <i>Dr. Sanni Kehinde Oseni — Rector, Gateway (ICT) Polytechnic</i>
10:40	Chief Guest of Honour Remarks <i>Engr. Dr. Kadir Kamoru Oluwatoyin, FNSE — Rector, The Federal Polytechnic, Offa</i>
10:50	Keynote Address 1 <i>Prof. Emeritus Alireza Baghai-Wadji — University of Cape Town, South Africa</i>
11:20	Keynote Address 2 <i>Prof. Y. L. Shuaib-Babata — University of Ilorin, Nigeria</i>
11:50	Group Photograph/Break
12:00	Keynote Address 3 <i>Dr. Arc. Oluseyi Israel Abatan — Moshood Abiola Polytechnic, Nigeria</i>
12:15	Keynote Address 4 <i>Dr.(Mrs) Adedoyin Mary A. — Lagos State University</i>
12:45	Industrial Lead Paper Presentation <i>Engr. Felix Adewale Odeleye — Nuclear Power Construction Engineer, UK</i>
13:15	Lunch Break
14:15	Parallel Technical Sessions Begin

DAY 3 — WEDNESDAY, 22 JULY

TIME	SESSION
10:30	Keynote Address 5 <i>Dr. Rilwan Kayode Apalowo — Universiti Sains Malaysia</i>
11:00	Keynote Address 6 <i>Dr. Emmanuel S. Babatunde — Achievers University, Owo</i>
11:30	Keynote Address 7 <i>Dr. Adeyemi Olusola — York University, Canada</i>
12:00	Selected Speaker <i>Adebayo Adeoti — University, Lubbock, Texas, USA</i>
	Parallel Technical Sessions and Poster Session continue thereafter

Parallel Technical Sessions and Poster Session continue thereafter.

Full session-by-session paper presentation schedules and timings for each day are published on the official conference website: <https://csetgaposa.github.io/icseecr2026/#program>

DAY 4 — THURSDAY, 23 JULY

Parallel Technical Sessions conclude, followed by the Closing Ceremony and Awards.

DAY 5 — FRIDAY, 24 JULY: DEPARTURE

Check-out and departure of delegates. No sessions scheduled.

TECHNICAL TRACK T-01

RENEWABLE ENERGY & SMART GRIDS

Accepted Abstracts

Security and Privacy Challenges in Cloud-Based Solar Panel Image Monitoring Systems

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Abstract—The exponential growth of cloud-connected solar photovoltaic (PV) monitoring systems has introduced a complex, evolving, and frequently underestimated landscape of security and privacy challenges that threaten the reliability and commercial viability of cloud-based solar monitoring infrastructure. This paper presents a systematic quantitative review of fifty (50) peer-reviewed academic studies published between 2018 and 2024, examining the principal security threats — including data breaches, man-in-the-middle attacks, privacy violations, denial-of-service attacks, malware injection, and insider threats — alongside the most widely proposed mitigation techniques spanning encryption, machine learning-based intrusion detection, blockchain, federated learning, and zero-trust architecture. Findings underscore that while encryption remains the most commonly adopted countermeasure, adversarial attacks on AI inference models and the absence of lightweight cryptographic standards for resource-constrained edge devices represent the most critical unresolved vulnerabilities. A novel five-layer security framework is proposed to address the unique constraints of IoT-to-cloud solar PV monitoring pipelines.

Index Terms—cloud security; solar PV; IoT; privacy; adversarial attacks; blockchain; federated learning; zero-trust.

Submission ID: ICSEECR-2026-090 | Track: T-01 | Presentation: In-Person | Oral

AI-Driven Classification of Solar Panel Defects: Healthy, Cracked, Delaminated, and Bird-Soiled Panels

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Abstract—The rapid global expansion of solar photovoltaic (PV) energy infrastructure has created an urgent need for intelligent, scalable, and cost-effective automated inspection systems capable of identifying physical panel degradation before it escalates into significant energy losses or system failures. Four principal defect categories — healthy (no defect), cracked (micro or macro cell fractures), delaminated (encapsulant layer separation), and bird-soiled panels (biological fouling) — each reduce photovoltaic conversion efficiency by measurable margins and require distinct remediation strategies. Manual inspection methods are inadequate at the scale of modern utility-scale solar farms, motivating the application of artificial intelligence (AI) and deep learning. This paper presents a rigorous quantitative systematic review of fifty (50) peer-reviewed research articles published between 2018 and 2024, drawn from IEEE Xplore, ScienceDirect, Scopus, and SpringerLink. The review systematically evaluates algorithm frequency, classification accuracy, dataset characteristics, hardware sensing platforms, application domains, recurrent limitations, and unresolved research gaps across state-of-the-art AI and machine learning methods for PV defect detection. Based on synthesis of these findings, a Hybrid CNN-SVM framework — combining a fine-tuned ResNet-50 feature extractor with a multi-class Support Vector Machine classifier — is proposed for simultaneous four-class defect classification. The proposed system achieves 97.4% overall accuracy and 97.3% macro F1-score on a 1,280-image four-class test set, outperforming all five baseline models including standalone ResNet-50 (95.1%) and VGG-16 (93.8%). Eight critical research gaps are identified and recommendations for future dataset development, edge deployment, and explainable AI integration are provided.

Index Terms—Solar photovoltaic defect classification; Convolutional neural networks; Transfer learning; Crack detection; Delamination; Bird-soiling; Deep learning; Edge AI; Systematic review; Support vector machine.

Submission ID: ICSEECR-2026-082 | Track: T-01 | Presentation: In-Person | Oral

Data Augmentation Techniques for Enhancing Machine Learning Models in PV Panel Fault Detection

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Abstract—The rapid global deployment of solar photovoltaic (PV) systems — with cumulative installed capacity surpassing 1.6 terawatts as of 2024 — has created an urgent need for reliable automated fault detection systems (International Renewable Energy Agency, 2024). Machine learning models are fundamentally constrained by the scarcity of large, well-annotated PV fault datasets and severe class imbalance between common fault types and rare but critical ones such as delamination and bypass diode failure. This paper presents a comprehensive investigation of data augmentation techniques for enhancing ML models in PV panel fault detection, grounded in a systematic quantitative review of fifty (50) peer-reviewed research articles published between 2018 and 2024. The reviewed studies span the full spectrum from classical geometric and photometric transformations to sophisticated generative methods including Generative Adversarial Networks (GANs), Denoising Diffusion Probabilistic Models (DDPMs), and foundation model-guided synthesis. Key findings establish that GAN-based augmentation delivers the highest accuracy improvement (+6.8 percentage points average), while diffusion model approaches achieve the highest image fidelity (FID scores as low as 9.8). A seven-layer Augmentation-Enhanced Machine Learning Pipeline (AEMLP) framework is proposed.

Index Terms—data augmentation; GAN; diffusion model; solar PV; fault detection; class imbalance; machine learning.

Submission ID: ICSEECR-2026-102 | Track: T-01 |
Presentation: In-Person | Oral

Spatial Pattern and Residents' Perception of Green Infrastructure Across Residential Densities in Abeokuta Metropolis, Ogun State, Nigeria

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Abstract—The unequal disproportion of urban green infrastructure (GI) across residential densities is a growing concern in planning and environmental justice, especially in fast growing cities in Africa. The rate of urban expansion has always been much faster than the

ability of the institute to incorporate GI into the planning process. The subject significantly aligns with the Sustainable Development Goal 11.7 which emphasizes sustainable cities and climate resilience goals. This goal explicitly commits to providing universal access to safe, inclusive, and accessible green and public spaces by 2030. In view of this, the study examined the spatial pattern of green infrastructure, and peoples' perceptions on the advantages of GI and its determinants and constraints in the three residential densities in Abeokuta Metropolis, Ogun State, Nigeria. Using a multi-stage sampling procedure that combined purposive neighbourhood selection within the local government, 22,882 of residential buildings were compiled as sample frame. A structured questionnaires of 326 were administered with respondents drawn from 61 neighbourhoods across high-density (n = 160), medium-density (n = 108), and low-density (n = 58) areas, while Kreijcie and Morgan (1970) formula was used to determine the sample size. The final stage applied systematic random sampling to select respondents within sampled households. Therefore, out of 326 questionnaires administered, 305 questionnaires retrieved formed the basis for the analysis of this study. Data were analysed using frequency distribution, percentages, one-way analysis of variance, and a Residents' Perception Index expressed as the Average Green Infrastructure Importance Index (AGIII) as a Resident's Perception Index which was determined by dividing Total Weight Value by the sample size of each residential density. Results showed that trees were the most prevalent dwelling-level GI (76.5%), while green walls were the least available (16.6%). The proportion of low-density areas with formal GI averaged 70.8%, compared to 22.7% for high-density areas. This difference across densities was significantly different across densities ($F = 15.256$, $p < 0.001$). Awareness of GI benefits also varied significantly across densities ($F = 31.817$, $p < 0.001$), with low-density respondents recording the highest AGIII (4.75) compared with medium- and high-density values of 3.94 and 3.81 respectively. The key leading factors in promoting GI availability were community and residents efforts (65.8%) and natural environmental features (55.4%), whereas poor maintenance (61.1%), lack of awareness (53.8%) and inadequate funding (46.9%) were the main challenges. The results indicate that, in Abeokuta, GI provision is structured by residential density and that comprehensive planning interventions should include density-specific GI approaches, continued investment in maintenance, and better community-government collaborations for equitable access to ecosystem services. Keywords: green infrastructure, ecosystem services, residential density, residents' perception, urban planning, Abeokuta

Index Terms—Green Infrastructure; Ecosystem Services; Residential Density; Residents' Perception; Urban Planning; Abeokuta.

Submission ID: ICSEECR-2026-029 | Track: T-01 |
Presentation: Oral

Development of a Dataset Management and Annotation System for Solar Faults Classification

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Abstract—Reliable fault detection and classification are essential for maintaining the operational efficiency and structural integrity of solar photovoltaic (PV) systems. A fundamental bottleneck constraining the performance of AI-based fault classification is the quality, completeness, and diversity of training datasets. This paper presents a systematic review of fifty (50) recent studies examining AI-based fault detection methodologies and associated dataset management challenges. The reviewed literature predominantly centres on Deep Learning (DL) and Machine Learning (ML), with CNNs (VGG16, ResNet, DenseNet) emerging as the standard for image-based defect detection. Cutting-edge architectures including Vision Transformers (ViT) and Kolmogorov-Arnold Networks (KAN) advance generalisation in small-sample scenarios. Multiple models achieve classification accuracies between 94% and 99.9%. The paper identifies critical dataset management gaps — annotation inconsistency, class imbalance, dataset silos, and the absence of standardised benchmarks — and proposes a comprehensive AI-driven dataset management and annotation system incorporating active learning, synthetic data generation, and federated data governance.

Index Terms—dataset management; annotation; solar fault classification; CNN; ViT; active learning; federated learning.

Submission ID: ICSEECR-2026-103 | Track: T-01 | Presentation: In-Person | Oral

Deep Learning Approaches for Automated Solar Panel Defect Classification

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Abstract—The global transition toward renewable energy has positioned solar photovoltaics (PV) as a critical pillar of sustainable power generation. Traditional manual inspection methods for solar panels are labour-intensive, cost-prohibitive, and prone to human error, especially as modern utility-scale solar farms encompass millions of individual cells. This paper explores the AJAYI OLUWAFUNKE AJOKE and application of deep learning (DL) methodologies for the

automated classification of solar panel defects, synthesising findings from 50 seminal research articles published between 2021 and 2026. The report highlights the efficacy of convolutional neural networks (CNNs) such as DenseNet169, ResNet50, and the YOLO family in detecting, localising, and classifying defects from multi-modal imaging sources including RGB, Infrared (IR) Thermography, and Electroluminescence (EL). The study addresses current limitations such as data scarcity and environmental noise and proposes future research directions toward zero-human-intervention diagnostic systems.

Index Terms—deep learning; solar panel; defect classification; CNN; YOLO; electroluminescence; thermal imaging.

Submission ID: ICSEECR-2026-088 | Track: T-01 | Presentation: In-Person | Oral

A Development of an Intelligent and Energy Efficient of a Brush Cutter with Sensor-Based Automatic Adjustable D.C Motor Speed Control (Automation and Feedback Focus)

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Abstract—People need ways to take care of farms and landscapes. This is why there is a demand for farming tools that do not waste energy. This study is about creating a brush cutter that can adjust its speed automatically. It uses sensors to figure out how dense the plants are and changes its speed accordingly. Traditional brush cutters always run at the same speed. This wastes a lot of energy, puts stress on the motor, and makes the user tired. The issue with these brush cutters is that they are unable to adapt to varying workloads. This indicates they have a short lifespan and are harmful to the environment. The goal of this study is to make a brush cutter that can change its speed automatically using real-time information from sensors. The system aims to create a control system for the motor speed, make it more energy efficient, and make it more comfortable for the user. We think that if the brush cutter can adjust its speed automatically, it will use energy and still cut plants effectively. The approach is to use signal processing and microcontrollers to control the motor speed. The sensors detect changes in resistance and torque while the brush cutter is working. They send this information to a control unit that adjusts the motor speed. The controller changes the speed based on how the brush cutter is working. We tested the prototype using plant densities to see how well it worked. We looked at how energy was used, how consistent the speed was, how much noise it made, and how hot it got. The smart brush cutter worked better than the others. This study is important because it combines automation, renewable energy, and smart control systems to make small-scale farming equipment. The

design also saves energy. The smart brush cutter is an example of how technology can make farming easier and more efficient. The main idea of this study is to make farming easier and more efficient by using brush cutters. Smart brush cutters like this one can really help people who take care of farms and landscapes. They can make the work easier and more efficient. This is why we need smart farming tools like this one. We hope that this study will encourage people to make smart farming tools. Smart brush cutters can make a difference in how we take care of farms and landscapes. They can help us use energy and make the work easier. This is why we think that smart brush cutters are the future of farming.

Index Terms—Intelligent Brush Cutter; Energy Efficiency; Sensor-Based DC Motor Control; Automatic Speed Regulation; Feedback Automation; Smart Agricultural Equipment.

Submission ID: ICSEECR-2026-131 | Track: T-01 |
Presentation: In-Person | Oral

footprint associated with battery production, disposal, and resource extraction. By facilitating improved battery maintenance and operational safety, the system supports the broader adoption of electric vehicles, a key technology for greenhouse gas emission reduction and sustainable transportation (Jafari et al., 2022; Mishra et al., 2021; Long et al., 2025). In summary, the developed SOH monitoring approach delivers a practical, reliable, and affordable solution suitable for application in EV battery management systems, renewable energy storage, and academic research, ultimately contributing to the advancement of sustainable electric mobility.

Index Terms—Battery State of Health (SOH); Lithium-Ion Battery; Electric Vehicles (EV); Battery Management System (BMS); Coulomb Counting; Internal Resistance Measurement; Embedded Microcontroller-Based Monitoring.

Submission ID: ICSEECR-2026-153 | Track: T-01 |
Presentation: In-Person | Oral

State of Health of Lithium Battery

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Abstract—The rising adoption of electric vehicles (EVs) has intensified the demand for efficient and reliable battery monitoring systems to ensure safety, longevity, and optimal performance of lithium-ion batteries. Among critical parameters, the State of Health (SOH) is a vital indicator reflecting battery aging, remaining useful life, and operational efficacy. This study presents the design, implementation, and experimental evaluation of a cost-effective and portable Battery SOH monitoring system tailored for lithium-ion batteries in EV applications. The system employs an embedded microcontroller (Arduino/STM32) interfaced with a voltage divider for voltage measurement, an ACS712 Hall-effect sensor for current sensing, and an NTC thermistor for temperature monitoring. Battery health estimation integrates Coulomb Counting for capacity assessment with DC internal resistance measurement, combined via a weighted SOH algorithm to enhance estimation accuracy and robustness under varying operational conditions. Real-time measurement of voltage, current, temperature, and computed SOH values are displayed on a 16×2 LCD, enabling continuous battery status monitoring without additional diagnostic tools. Experimental validation on a four-cell lithium-ion battery module subjected to accelerated cycling demonstrates measurement accuracies of approximately ± 0.1 V (voltage), ± 0.5 A (current), and ± 2 °C (temperature). The SOH estimation deviated by less than $\pm 4\%$ compared to reference metrics, while internal resistance progressively increased from 85 mΩ to 102 mΩ, consistent with normal battery aging trends. Notably, this monitoring system aligns with climate and sustainability objectives by promoting prolonged battery lifespan and optimized utilization, thereby reducing battery replacements and lowering the environmental

Comparative Analysis of RGB and Thermal Imaging in Solar Panel Fault Diagnosis

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Abstract—This paper presents a systematic comparative analysis of RGB and thermal imaging technologies for solar panel fault diagnosis, drawing on a structured review of fifty (50) peer-reviewed research articles published between 2018 and 2024. RGB imaging achieves a mean classification accuracy of approximately 95.6% across reviewed studies, with peak performance of 99.1% demonstrated by transformer-based architectures. Thermal imaging achieves a comparable mean accuracy of 95.2%, with peak performance of 98.7% using Vision Transformer models. RGB imaging excels for surface-visible defects such as micro-cracks (91% average detection) and soiling, while thermal imaging demonstrates superior performance for hotspots (96%), bypass diode failures, and delamination. A dual-modality fusion approach achieves an accuracy of 99.3%, representing the optimal direction for future research and deployment.

Index Terms—RGB imaging; thermal infrared imaging; PV fault diagnosis; CNN; dual-modality fusion; hotspot detection; solar panel.

Submission ID: ICSEECR-2026-092 | Track: T-01 |
Presentation: In-Person | Oral

IoT-Integrated Monitoring of Photovoltaic Panels Using Computer Vision Algorithms

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Abstract—The rapid expansion of solar photovoltaic (PV) installations globally has created a critical need for reliable, automated monitoring systems capable of identifying faults before significant energy losses occur. This paper presents a quantitative analysis of fifty (50) peer-reviewed research articles published between 2019 and 2024, examining the application of computer vision and machine learning algorithms within Internet of Things (IoT) frameworks for PV panel monitoring. Articles were systematically selected from IEEE Xplore, ScienceDirect, and Springer databases using a PRISMA-style screening protocol. Key findings indicate that convolutional neural networks (CNN) dominate the literature, representing 32% of reviewed studies with a mean classification accuracy of 96.1%. Hybrid CNN-SVM architectures achieve the highest combined performance at an average of 97.8% accuracy. YOLOv5 and YOLOv8 models are increasingly favoured for real-time edge deployment, recording a mean accuracy of 94.8%. Eight key research gaps are identified, and an optimised IoT-integrated hybrid CNN-SVM framework for continuous real-time PV monitoring is proposed.

Index Terms—IoT; photovoltaic monitoring; CNN; YOLO; computer vision; fault detection; hybrid CNN-SVM.

Submission ID: ICSEECR-2026-087 | Track: T-01 | Presentation: In-Person | Oral

Land Grabbing, Inequality and Corporate/Financial Land Concentration in the Economy

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Abstract—Land grabbing has become a core issue in the contemporary politics of land, which over the past few years has gained momentum in Africa, Asia, and Latin America, with states, multinational corporations, agribusiness conglomerates, and financial actors buying up massive areas of land for commercial agriculture, extractive industries, infrastructural development, and speculative investment. Land grabbing has emerged as a defining feature of contemporary global land politics, driven by the growing involvement of multinational corporations, agribusiness conglomerates and financial actors such as private equity funds, pension schemes and

sovereign wealth funds. This phenomenon extends beyond the traditional dynamics of territorial dispossession and now reflects a deeper structural transformation in which land is increasingly treated as a financial asset rather than a socio-economic resource. As corporations and financial institutions consolidate control over vast expanses of agricultural and resource-rich land, patterns of ownership become heavily concentrated, producing and intensifying multiple layers of inequality. Economic inequality arises as land, a critical factor of production, shifts from smallholder and communal ownership to large-scale commercial estates that marginalize rural livelihoods. Social and gender inequality deepen as vulnerable groups, particularly women and indigenous communities, face displacement, loss of access to common resources and weakened bargaining power. Political inequality also expands, as state – corporate alliances undermine democratic land governance and erode community participation. The resulting corporate and financial land concentration reinforces an unequal agrarian structure in which profit maximization and speculative investment overshadow local development priorities. Consequently, land grabbing not only redistributes land upward but also restructures power relations in ways that entrench poverty, dispossession and exclusion. Understanding these linkages is essential for designing equitable land governance frameworks capable of addressing the interconnected challenges of land grabbing, inequality and corporate concentration

Submission ID: ICSEECR-2026-095 | Track: T-01 | Presentation: In-Person | Oral

Development of Embedded Systems for on-Site Photovoltaic Panel Health Assessment: Combining Microcontrollers with Image Capture and Processing for Portable Diagnostics

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Abstract—The rapid global expansion of solar photovoltaic (PV) energy necessitates efficient, on-site health assessment of PV panels to mitigate power loss and ensure long-term system reliability. However, traditional diagnostic methods manual visual inspection, laboratory-based electroluminescence imaging, and I-V curve tracing are characterised by operational downtime requirements, high cost, and the absence of real-time portability, rendering them inadequate for large-scale solar farm deployments. This paper presents a systematic quantitative review and comparative analysis of fifty (50) high-impact peer-reviewed articles sourced from IEEE Xplore, ScienceDirect, and SpringerLink (2016 – 2025), focused on embedded system approaches to on-

site PV panel health assessment. The review systematically evaluates hardware architectures, machine learning algorithms, dataset characteristics, application domains, recurring limitations, and unresolved research gaps. Key findings indicate that microcontroller-based systems utilising optimised Convolutional Neural Networks (CNNs) achieve defect detection accuracies of 92% – 99%, that fused thermal-RGB imaging techniques reduce false-positive rates by approximately 12.5%, and that edge-optimised TinyCNNs substantially reduce memory overhead without compromising diagnostic fidelity. Six critical research gaps are identified, including the absence of hybrid multi-modal embedded diagnostic systems, limited real-time edge deployment, and a complete absence of embedded cybersecurity frameworks. In response, a conceptual Hybrid Multi-Modal Digital Twin (HM-DT) framework is proposed, integrating low-power microcontrollers with real-time multi-modal image capture, TinyML-based edge inference, CNN-SVM feature fusion, digital twin synchronisation, and autonomous control. This framework provides a scalable, cost-efficient pathway for next-generation on-site PV health assessment, bridging the critical gap between high-fidelity laboratory diagnostics and practical, continuous field deployment.

Submission ID: ICSEECR-2026-085 | Track: T-01 |
Presentation: In-Person | Oral

An Assessment of the Role of Urban Green Spaces in Enhancing Climate Resilience in Ogun State

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Abstract—The study assesses the role of urban green spaces in enhancing climate resilience in Ogun state. Nigeria's urban regions are more susceptible to the effects of climate change, such as rising temperatures, flooding, erosion, and environmental degradation. Parks, gardens, urban forests, green belts, and recreational open spaces are examples of urban green spaces that are important for reducing these problems and fostering environmental sustainability. The Study covers the environmental professionals in Abeokuta due to rapid urban growth. In Ogun State, Nigeria, this study evaluates how urban green spaces can improve climate resilience. The sample size for the study is consists of 15 registered urban planners, 15 environmental officers, 15 Architects and 15 staffs of Ministry of Land and Buildings in the Abeokuta. Descriptive and Inferential statistics were employed to analyze the data. Descriptive statistical tools including frequencies, percentages, mean scores, and ranking techniques are used to examine data. The results showed that urban green areas greatly contribute to climate resilience. Despite these

advantages, their efficacy is constrained by elements including fast urbanization, shoddy planning, and encroachment on green spaces, insufficient funding, shoddy policy execution and subpar maintenance procedures. The study concludes that urban green spaces are essential for sustainable urban development and climate resilience in Ogun State. The Study recommends increased investment in green infrastructure, stronger environmental policies, effective urban planning, public awareness initiatives and improved management strategies to ensure the protection, expansion, and sustainable use of urban green spaces for present and future generations.

Submission ID: ICSEECR-2026-109 | Track: T-01 |
Presentation: In-Person | Oral

Optimisation of Computer Vision Algorithms for Efficient Solar Panel Fault Detection

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Abstract—The global deployment of photovoltaic (PV) solar systems has reached unprecedented scale, with over 1.6 terawatts of cumulative installed capacity as of 2024 (International Renewable Energy Agency, 2024). Automated fault detection using computer vision (CV) algorithms is critical for maintaining system performance, yet deployment is frequently hindered by computational constraints, cross-domain generalisation failures, and the absence of systematic optimisation frameworks. This paper presents a comprehensive investigation into the optimisation of CV algorithms for efficient solar panel fault detection, grounded in a systematic quantitative review of fifty (50) peer-reviewed articles published between 2018 and 2025. The reviewed studies encompass the full spectrum from classical SVM methods to state-of-the-art foundation model adaptations, with a focus on model pruning, quantisation, knowledge distillation, transfer learning, neural architecture search (NAS), and real-time inference optimisation. Key findings reveal that transfer learning is the dominant optimisation strategy (28% of studies), while hybrid ensemble approaches achieve the highest mean accuracy (97.8%). Bayesian-optimised YOLOv8 and vision transformer models represent the current performance frontier with mean accuracies of 98.2 – 99.4%. Keywords: computer vision; solar PV; fault detection; model optimisation; YOLO; transfer learning; neural architecture search; edge deployment

Keywords —
Index Terms—Computer Vision; Solar PV; Fault Detection; Model Optimisation; YOLO; Transfer Learning; Neural Architecture Search.

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Presentation: In-Person | Oral

Implementation of a Web-Based E-Learning Platform with Interactive AI-Powered Content and Performance Tracking.

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Abstract—The proliferation of artificial intelligence (AI) technologies has transformed digital education, yet most web-based learning platforms still rely on static content delivery mechanisms. This paper presents a quantitative analysis of 60 peer-reviewed studies (2014 – 2025) investigating AI-powered content delivery, adaptive learning, and performance tracking through a PRISMA-guided systematic review across IEEE Xplore, ScienceDirect, Springer, arXiv, and Google Scholar. The results show that adaptive and recommendation algorithms dominate the field (35.0%), followed by real-time analytics (21.7%) and large language models (20.0%). Web-based architectures account for 73.3% of the platforms reviewed, while only seven studies report quantitative performance metrics, with classification accuracy ranging from 94.0% to 98.2%. Nine evidence-based research gaps are identified. Based on these findings, a hybrid AI-powered e-learning framework is proposed, integrating adaptive content delivery, GPT-based tutoring, real-time performance analytics, gamification, and privacy-by-design data governance.

Index Terms—e-learning; adaptive learning; large language models; intelligent tutoring; performance tracking; PRISMA; learning analytics; GPT-4; federated learning.

Submission ID: ICSEECR-2026-078 | Track: T-01 |
Presentation: In-Person | Oral

Design and Implementation of an AI-Based Face Recognition System for Student Security and Access Control

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Abstract— The growing need for improved security, identity verification, and access management within educational institutions has increased the adoption of biometric technologies as reliable alternatives to traditional authentication methods. Conventional security mechanisms such as identity cards, passwords, and manual verification procedures are often vulnerable to loss, theft, duplication, impersonation, and human

error, particularly in large student populations where manual verification becomes impractical. While facial recognition has been widely studied for general security applications, existing literature on AI-based access control in academic institutions often focuses narrowly on either detection accuracy or database management in isolation, with limited attention to an integrated, end-to-end pipeline spanning image acquisition, preprocessing, CNN-based classification, and secure database verification within a single deployable framework tailored to institutional access control. This paper addresses that gap by presenting the design and implementation of a complete AI-based face recognition system developed specifically to enhance student security and access control within academic environments. The system was developed through a structured four-stage methodology. First, requirements analysis identified core functional needs, including real-time identification, resistance to impersonation, and integration with existing student records. Second, in the image acquisition and preprocessing stage, facial images were captured via a digital imaging device and processed through normalization, feature enhancement, resizing, and noise reduction to improve recognition performance under varying conditions. Third, a Convolutional Neural Network (CNN)-based architecture was designed and trained for automatic feature extraction and facial classification, with the training dataset curated to represent varying illumination conditions, facial expressions, and head orientations to improve model robustness. Fourth, system integration and testing involved connecting the recognition module to a secure database storing biometric templates and student records, followed by performance evaluation using accuracy, precision, recall, response time, and reliability as evaluation criteria. Experimental results demonstrate that the developed system achieves high recognition accuracy while significantly reducing the possibility of unauthorized access and identity impersonation compared to conventional card- or password-based methods. Beyond security enhancement, the system's architecture supports extension to attendance management, examination verification, laboratory access control, and smart campus administration. This work contributes to institutional digital security infrastructure and aligns with SDG 4 (Quality Education) by supporting safer, more reliable learning environments through robust access control, and SDG 9 (Industry, Innovation, and Infrastructure) by demonstrating a scalable, AI-driven framework adaptable to Internet of Things (IoT)-enabled smart educational infrastructures. The study provides a practical, reproducible reference for institutions seeking cost-effective, intelligent biometric security solutions. Keywords: Face Recognition, Biometrics, Student Security, Access Control, Artificial Intelligence, Deep Learning, Convolutional Neural Network.

Index Terms—Face Recognition; Biometrics; Student Security; Access Control; Artificial Intelligence; Deep Learning; Convolutional Neural Network.

Submission ID: ICSEECR-2026-148 | Track: T-01 |
Presentation: Virtual | Oral

Artificial Intelligence-Based Image Analysis Techniques for Photovoltaic Panel Inspection: A Comprehensive Review

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Abstract— The rapid growth of photovoltaic (PV) installations worldwide has created an increasing demand for intelligent monitoring and maintenance systems capable of ensuring optimal energy production, operational reliability, and extended service life. Traditional inspection methods are often labor-intensive, time-consuming, expensive, and prone to human error, particularly in large-scale solar farms. Consequently, Artificial Intelligence (AI)-based image analysis has emerged as a transformative technology for automated photovoltaic panel inspection, fault detection, and predictive maintenance. While numerous studies have investigated individual imaging modalities or specific deep learning architectures for PV fault detection, few reviews have systematically compared imaging modalities, algorithmic approaches, and deployment platforms (UAVs, robotics, cloud systems) within a single unified framework. Most existing reviews focus narrowly on either the imaging technique or the classification algorithm in isolation, without addressing how these components interact across the full inspection pipeline, from data acquisition to real-time deployment. This review addresses that gap by providing an integrated, criteria-based synthesis spanning imaging modalities, AI architectures, defect categories, and deployment platforms, evaluated collectively rather than as disconnected strands of research. This review follows a structured selection methodology. Literature published between 2016 and 2025 was retrieved from IEEE Xplore, ScienceDirect, MDPI, and Google Scholar using search terms including "PV fault detection," "solar panel image analysis," "deep learning photovoltaic inspection," and "UAV thermal imaging PV." Retrieved studies were screened for methodological clarity, reported quantitative performance data, and relevance to photovoltaic-specific defect detection, then organized by imaging modality (infrared thermography, electroluminescence, ultraviolet, hyperspectral, RGB imaging) and algorithmic approach (Convolutional Neural Networks, transfer learning models, object detection frameworks, recurrent neural networks, hybrid architectures). Each approach was assessed against fault classification accuracy, localization capability, computational demand, and suitability for real-time deployment. Particular attention is given to detection of hotspots, microcracks, delamination, soiling, shading effects, cell degradation, and electrical abnormalities that degrade energy output. The integration of unmanned

aerial vehicles (UAVs), autonomous robotic platforms, and cloud-based monitoring systems for large-scale data acquisition is also examined. Comparative analysis reveals that deep learning-based approaches consistently outperform conventional image processing techniques in accuracy, robustness, and scalability, though challenges related to dataset availability, environmental variability, and model generalization remain unresolved. By consolidating dispersed technical findings into a unified diagnostic framework, this review supports SDG 7 (Affordable and Clean Energy) by enabling more efficient, cost-effective PV maintenance, and SDG 13 (Climate Action) by improving the reliability of solar infrastructure central to decarbonization efforts. The findings offer researchers and practitioners a consolidated reference for developing next-generation AI-powered diagnostic frameworks for renewable energy applications. Keywords: Artificial Intelligence, Photovoltaic Panels, Image Analysis, Deep Learning, Fault Detection, Predictive Maintenance.

Index Terms—Artificial Intelligence; Photovoltaic Panels; Image Analysis; Deep Learning; Fault Detection; Predictive Maintenance.

Submission ID: ICSEECR-2026-149 | Track: T-01 | Presentation: Virtual | Oral

A Comprehensive Review of Optimization Techniques for Enhancing Photovoltaic Power System Performance

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Abstract— The increasing global demand for sustainable and environmentally friendly energy sources has accelerated the deployment of photovoltaic (PV) systems as viable alternatives to conventional fossil-fuel-based electricity generation. Despite continuous improvements in photovoltaic technologies, the performance and efficiency of PV systems remain significantly affected by environmental and operational factors such as fluctuating solar irradiance, temperature variations, dust accumulation, aging effects, and partial shading conditions. These challenges often result in reduced power output and lower system efficiency, necessitating optimization techniques to maximize energy extraction and improve operational reliability. While several reviews have examined Maximum Power Point Tracking (MPPT) methods individually or compared a narrow subset of conventional and intelligent techniques, few have systematically synthesized conventional, intelligent, and hybrid optimization approaches within a single comparative framework that also addresses emerging IoT-enabled and predictive optimization trends. This review fills that gap by providing an integrated, criteria-based comparison of

optimization techniques spanning three generations of MPPT development, evaluated against a consistent set of performance metrics rather than treated as isolated case studies. This paper adopts a structured review methodology. Relevant literature published between 2015 and 2025 was retrieved from databases including IEEE Xplore, ScienceDirect, and Google Scholar, using search terms such as "MPPT optimization," "photovoltaic performance," "hybrid optimization algorithms," and "AI-based PV control." Papers were screened for relevance, methodological clarity, and reported quantitative performance data, then categorized into conventional MPPT methods (Perturb and Observe, Incremental Conductance, Hill-Climbing, Constant Voltage), intelligent approaches (Artificial Neural Networks, Fuzzy Logic Controllers, Genetic Algorithms, Particle Swarm Optimization, Ant Colony Optimization), and hybrid frameworks. Each technique was evaluated against six criteria: tracking accuracy, convergence speed, computational complexity, adaptability to environmental variation, implementation requirements, and overall impact on energy yield. Comparative analysis indicates that conventional methods such as Perturb and Observe and Incremental Conductance typically achieve tracking efficiencies of 92 – 96% under steady-state irradiance but degrade markedly, often below 85%, under rapidly changing or partial shading conditions due to oscillation around the maximum power point. Intelligent techniques, including Fuzzy Logic Controllers and Particle Swarm Optimization, report tracking efficiencies of 97 – 99.5% with convergence times reduced by 30 – 60% relative to conventional methods, though at the cost of higher processor and memory requirements that raise per-unit hardware expenses and complicate deployment in low-cost residential systems. Hybrid frameworks combining intelligent algorithms with conventional logic (e.g., Fuzzy-PSO, ANN-Incremental Conductance) report further gains, with several studies documenting energy yield improvements of 5 – 15% over standalone methods under partial shading, but these gains come with increased tuning complexity, greater sensitivity to training data quality, and longer development cycles. Emerging IoT-enabled and predictive optimization approaches integrate real-time sensor networks (irradiance, temperature, and panel-level voltage/current monitoring) with cloud- or edge-based predictive models, most commonly employing recurrent neural networks or ensemble learning, to forecast irradiance shifts and preemptively adjust duty cycles before power loss occurs; reviewed studies report additional yield gains of 2 – 8% over reactive MPPT alone, though at the cost of communication latency, data security concerns, and infrastructure investment that limit near-term scalability in resource-constrained settings. By consolidating dispersed findings into a single comparative framework, this review directly supports SDG 7 (Affordable and Clean Energy) by identifying cost-effective pathways to improve PV efficiency, and SDG 13 (Climate Action) by advancing optimization strategies that reduce reliance on fossil-fuel-based generation. The review offers researchers, engineers, and

policymakers a consolidated reference for selecting efficient, cost-effective optimization strategies to support the global transition toward sustainable energy systems. Keywords: Photovoltaic Systems, Maximum Power Point Tracking, Optimization Techniques, Artificial Intelligence, Renewable Energy, Particle Swarm Optimization.

Index Terms—Photovoltaic Systems; Maximum Power Point Tracking; Optimization Techniques; Artificial Intelligence; Renewable Energy; Particle Swarm Optimization.

Submission ID: ICSEECR-2026-154 | Track: T-01 |
Presentation: Virtual | Oral

Development of a Smart Agricultural System Using IoT Sensors and Data Analytics

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Abstract— Smart agriculture is transforming conventional Agricultural methods toward sustainable crop management. It is driven by developments in the Internet of Things (IoT), artificial Intelligence (AI), and data analytics. An overview of the main Technologies enabling smart agriculture is given in this paper, along with examples of how they are being used to optimize certain crop farming characteristics. Food Security is under a growing threat of climate change with the increasing temperatures, the erratic rainfall patterns, and the extreme weather events that are disrupting agricultural productivity and endangering food supplies. The Food and Agriculture Organization (FAO) estimates that world agriculture could lose yields by no more than 2%. An increase of 6% by 2050 is expected. The food security for the growing global population will be ensured only when the agricultural practices are adaptive and resilient. In order to deliver useful insights for crop disease Diagnosis, pest management, and yield prediction, artificial Intelligence (AI) algorithms evaluate the massive volume of data gathered from these sensors. In this regard, digital agriculture is gaining a lot of momentum. It includes IoT, big data analytics, AI, and remote sensing technologies, which are critical for transforming agriculture in terms of how it is controlled. Digital tools that assist in precision agriculture make it possible to use resources more efficiently, as they optimizes the irrigation, fertilization, and pest management. This paper presents a systematic review of smart agricultural systems that integrate Internet of Things (IoT) sensors and data analytics to improve crop monitoring, soil management, and irrigation practices. A

modified PRISMA-guided selection process identified 50 peer-reviewed articles published between 2022 and 2026 from IEEE Xplore, ScienceDirect, Springer, ArXiv/Zenodo, and Google Scholar. Findings indicate that machine learning models — particularly Convolutional Neural Networks (CNN), hybrid ensemble approaches, Gradient Boosting, and TinyML — achieve classification and prediction accuracy rates above 90%, with some models exceeding 99%. Smart irrigation systems consistently report water-use reductions of 20 – 50% and crop yield improvements exceeding 20%. Scalable hierarchical edge – cloud architectures improve system responsiveness while minimising latency. The paper synthesises algorithmic performance (Table I – II), dataset characteristics (Table III), hardware deployment (Table IV), application distribution (Table V), identified limitations (Table VI), research gaps (Table VII), and annual research trends (Table VIII), and proposes a next-generation AIoT – Edge – Cloud framework for sustainable precision agriculture.

Index Terms—Smart Agriculture; IoT Sensors; Precision Farming; Soil Health Monitoring; Data Analytics.

Submission ID: ICSEECR-2026-133 | Track: T-01 |
Presentation: Virtual | Oral

Design and Construction of a Solar Charging and Irradiance Monitoring System for Electric Vehicles

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Abstract— This paper presents the design and construction of a solar-powered charging and irradiance monitoring system for electric vehicles (EVs). The system integrates a photovoltaic (PV) array, a Perturb-and-Observe (P&O) MPPT charge controller, rechargeable battery storage, and an Arduino/ESP32 microcontroller with irradiance sensing and display. A MATLAB/Simulink simulation of a 63.94 kW PV array (30 parallel × 10 series panels at 950 W/m², 30°C) validated a DC-DC boost converter in discontinuous conduction mode (DCM) for pulsed EV battery charging. Hardware testing confirmed irradiance sensing within ±8% across 200 – 950 W/m² and correct SOC-controlled charging halting per Eq. (1), demonstrating solar energy as a viable sustainable EV charging infrastructure for developing regions. Intelligent fuzzy controller optimization for PV systems [60] is reviewed as a future enhancement path. Over the past few years, the increasing use of electric vehicles (EVs) due to rising CO₂ emissions, depletion of fossil fuels, and incentives for renewable energy sources. To facilitate the transition

to EVs, the growth of charging stations is crucial. Improvements in EV batteries and battery management are also linked to user experiences and governmental policies. The study focuses on a solar-powered smart EV charger that uses IoT technology to monitor the charging and discharging of batteries, prolong their lifespan, and reduce the total expense of EVs. The system comprises a solar panel, battery, and microcontroller that communicate with a smartphone app or web server. The system is useful for remote areas, camping, emergency situations, and environmental monitoring. The system can be customized to different user needs and preferences. The paper provides insights into the development of sustainable EV charging solutions. The research methodology includes the design and integration of hardware and software, sensor calibration, and both laboratory and field testing. The test results indicate that the system can provide accurate and consistent data with minimal measurement error. The monitoring system measures solar irradiation with an average deviation of less than 2% and records electrical parameters with high precision. The hardware and software design for the solar irradiation monitoring system and load distribution in grid-connected solar power systems was successfully implemented, integrating components such as the PZEM-004t sensors and pyranometer. The firmware played a critical role in ensuring seamless interaction between hardware and software, enabling real-time data collection and transmission to Firebase, with data transmission latency averaging 0.274 seconds. Both laboratory and field tests confirmed the system's accuracy, with the pyranometer demonstrating an accuracy of 97.97% and a sensitivity of 0.0814, showcasing its potential for effective solar energy monitoring. The findings of this research are expected to promote the optimal utilization of solar energy in the future, advancing both industrial innovation and sustainable resource management.

Index Terms—Solar Irradiance; EV Charging; Mppt; Photovoltaic System; Embedded Systems.

Submission ID: ICSEECR-2026-134 | Track: T-01 |
Presentation: Virtual | Oral

Image Processing-Based Fault Detection System for Photovoltaic Panel Using Python and OpenCV

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Abstract—The rapid global expansion of solar photovoltaic (PV) systems surpassing 1.6 TW of installed capacity in 2023 and projected to reach 5 TW by 2030 positions automated fault detection as a critical enabler of reliable clean energy delivery, directly supporting UN Sustainable Development Goal 7 (Affordable and Clean Energy) and SDG 13 (Climate

Action). Manual and periodic thermographic inspection methods are economically and logistically inadequate at the scale of modern utility solar farms, motivating a rigorous evidence synthesis of emerging automated alternatives. This paper presents a PRISMA 2020-compliant systematic literature review of 50 peer-reviewed studies published between 2015 and 2025. Candidate records were retrieved from five academic repositories IEEE Xplore, ScienceDirect, SpringerLink, Google Scholar, and ResearchGate using structured Boolean keyword strings targeting image processing, deep learning, and computer vision approaches to PV fault detection. An initial yield of 276 records was reduced through sequential deduplication, title-and-abstract screening, full-text eligibility assessment, and a quality threshold requiring clear methodology, reproducible experimental conditions, and quantitative or substantiated qualitative outcomes, yielding a final corpus of 50 studies. Structured data extraction captured algorithm architecture, sensing modality, dataset characteristics, evaluation protocol, and reported performance metrics for each included study, enabling systematic cross-study comparison. The review finds that deep learning approaches particularly convolutional neural networks (CNNs), YOLO-family object detectors, and Vision Transformers dominate the field (56% of studies), with leading architectures achieving detection accuracies exceeding 95% and inference speeds up to 280 FPS. Self-supervised contrastive learning achieves greater than 95% accuracy with a tenfold reduction in labelled training requirements, directly addressing the data scarcity challenge affecting 74% of reviewed studies that relied on private, non-reproducible datasets. Hybrid multi-modal frameworks integrating thermal infrared imagery, RGB visual data, and electrical I-V curve analysis demonstrate superior fault coverage over single-modality approaches. Persistent barriers to deployment include computational complexity on edge hardware, cross-domain generalisation failure exceeding 5% accuracy degradation across geographic domains, the absence of standardised public benchmark datasets, and the critical lack of quantitative linkage between visual defect localisation and watt-peak power losses a gap with direct consequences for O&M financial modelling and climate-aligned energy asset management. To address these gaps, a conceptual Multi-Modal Hybrid Edge-Transformer (MHE-T) framework is proposed, integrating feature-level cross-modal fusion across RGB, thermal IR, I-V curve, and environmental sensor inputs; a hybrid CNN-Transformer encoder with efficient long-range self-attention; INT8-quantised onboard edge inference targeting ≥ 25 FPS within a 15 W power budget; adversarial domain adaptation for cross-farm generalisation; and a power-loss regression head bridging visual fault detection to operational energy impact. Seven future research directions are identified, including a community-curated multi-modal PV Fault Benchmark dataset, federated learning for privacy-preserving cross-installation knowledge transfer, and longitudinal remaining-useful-life modelling. Collectively, these advances support climate resilience

objectives and the global transition to a carbon-neutral energy system.

Index Terms—solar photovoltaic; fault detection; systematic literature review; PRISMA; deep learning; convolutional neural networks; Vision Transformer; edge computing; multi-modal fusion; climate resilience; SDG 7; SDG 13.

Submission ID: ICSEECR-2026-084 | Track: T-01 |
Presentation: In-Person | Oral

A Hybrid Artificial Intelligence Framework for Predictive Maintenance and Grid Reliability Enhancement in Renewable-Energy-Integrated Power Systems

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Abstract—The global transition toward sustainable energy systems has accelerated the integration of renewable energy resources into modern power networks. While renewable energy technologies such as solar photovoltaic and wind power contribute significantly to decarbonization efforts, their increasing penetration introduces operational uncertainties that challenge the reliability, stability, and resilience of electrical power systems. In many developing economies, including Nigeria, these challenges are compounded by aging infrastructure, frequent equipment failures, limited maintenance resources, and growing electricity demand. Consequently, maintaining reliable power delivery while supporting renewable energy integration has emerged as a critical concern for utility operators and policymakers. Conventional maintenance practices in power systems are predominantly reactive or time-based, often resulting in delayed fault identification, unplanned outages, increased operational costs, and inefficient utilization of maintenance resources. Although significant advances have been made in smart grid monitoring and renewable energy forecasting, limited attention has been given to integrating equipment health assessment, renewable energy variability, and maintenance decision-making within a unified intelligent framework. This gap highlights the need for proactive and data-driven approaches capable of improving asset reliability in increasingly complex power system environments. This study proposes a hybrid Artificial Intelligence (AI) framework for predictive maintenance and grid reliability enhancement in renewable-energy-integrated power systems. The primary research objective is to develop a unified decision-support framework that integrates equipment health assessment and renewable energy variability analysis to improve maintenance planning, fault prediction, and operational reliability. The hybrid architecture combines Random Forest algorithms for asset condition assessment and fault classification with Long Short-Term Memory (LSTM)

networks for time-series prediction of equipment degradation and renewable generation variability. This hybrid approach is expected to provide superior predictive capability compared with single-model approaches by simultaneously capturing asset-condition characteristics and temporal operational patterns associated with renewable energy intermittency. The framework integrates real-time condition monitoring, Internet of Things (IoT)-enabled sensing technologies, and machine-learning-based failure prediction models to support risk-informed maintenance decisions. Asset health indicators considered include transformer oil temperature, dissolved gas analysis trends, loading conditions, voltage deviations, fault history, equipment age, and operational stress factors. Renewable generation variability metrics include power output fluctuations, forecast errors, ramp-rate variations, and intermittency indices associated with solar and wind generation. By continuously analysing operational and environmental data from critical grid components such as transformers, transmission lines, and distribution equipment, the proposed system enables early fault detection, prediction of equipment degradation, and optimisation of maintenance schedules. The framework will be validated through simulation studies using IEEE benchmark power system test networks and historical operational datasets, with performance assessed based on fault prediction accuracy, maintenance effectiveness, reliability improvement, and expected reduction in equipment downtime. The proposed approach is expected to enhance grid resilience by reducing unexpected equipment failures, minimising maintenance-related downtime, extending asset lifespan, and improving the integration of renewable energy resources into existing power networks. Furthermore, the framework supports the ongoing digital transformation of power systems by promoting intelligent asset management and operational efficiency. The study contributes to the development of sustainable and resilient energy infrastructure by providing a scalable solution for improving power system reliability in both developed and emerging economies. The findings are expected to offer valuable insights for researchers, utility companies, and policymakers seeking to modernise electricity networks and accelerate the transition toward smarter and more sustainable energy systems. **Keywords:** Artificial Intelligence, Predictive Maintenance, Smart Grid, Renewable Energy Integration, Grid Reliability, Internet of Things (IoT).

Index Terms—Artificial Intelligence; Predictive Maintenance; Smart Grid; Renewable Energy Integration; Grid Reliability; Internet Of Things (IoT).

Submission ID: ICSEECR-2026-044 | Track: T-01 |
Presentation: In-Person | Oral

RFID-Based Access Authentication System for Electric Vehicle Operation

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Abstract—The increasing adoption of electric vehicles (EVs) has created a growing need for improved security mechanisms that prevent unauthorized vehicle operation while maintaining convenience for legitimate users. Conventional mechanical keys and basic ignition systems are susceptible to theft, duplication, and unauthorized access, necessitating the development of more reliable authentication methods. This project presents the design and implementation of an RFID-based access authentication system for secure electric vehicle operation. The proposed system utilizes a high-frequency RC522 RFID reader operating at 13.56 MHz and compliant with the ISO/IEC 14443A standard, an Arduino Uno or ESP32 microcontroller as the processing unit, and a 5 V opto-isolated relay module that controls the motor enable circuit of the electric vehicle. During operation, the RFID reader detects the unique identifier (UID) of an RFID card or tag presented by the user. The microcontroller compares the received UID with a database of authorized identifiers stored in its EEPROM memory. When a valid match is detected, the relay is energized, allowing the vehicle motor to operate. Conversely, unauthorized or unregistered RFID tags are denied access, ensuring that the vehicle remains disabled. The system was designed with emphasis on reliability, low power consumption, rapid response, ease of implementation, and cost-effectiveness, making it suitable for educational, commercial, and personal electric vehicle applications. Experimental evaluation was conducted through fifty authentication cycles using both authorized and unauthorized RFID tags. The results demonstrated a 100% authentication success rate for registered users, an average authentication response time of approximately 420 milliseconds, and zero false activations throughout the testing period. The developed prototype significantly enhances vehicle security by eliminating conventional key-based vulnerabilities while providing a simple, contactless, and user-friendly means of vehicle authorization. Furthermore, the system offers flexibility for future enhancements such as encrypted RFID communication, mobile application integration, cloud-based access logging, GPS tracking, biometric verification, and Internet of Things (IoT) connectivity. The study demonstrates that RFID technology provides an efficient, affordable, and dependable solution for improving access control and operational security in modern electric vehicles.

Index Terms—RFID Authentication; Electric Vehicle Security; Access Control; Embedded Systems; RC522 RFID Reader; Contactless Authentication; Motor Enable System; Arduino; ESP32; Relay Control.

Submission ID: ICSEECR-2026-122 | Track: T-01 |
Presentation: In-Person | Oral

Waste to Energy Technologies in Ogun State: A Critical Review of Strategies for Addressing Energy and Waste Challenges

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Abstract—The increasing rate of urbanization, population growth, and industrial activities in Ogun State, Nigeria, has intensified challenges related to municipal solid waste management and energy supply. Waste-to-energy (WtE) technologies have emerged as viable solutions for simultaneously reducing waste volumes and generating sustainable energy. This review critically examines the potential, applicability, and implementation challenges of major Waste-to-energy technologies in Ogun State, including incineration, anaerobic digestion, landfill gas recovery, gasification, and pyrolysis. The study evaluates existing literature on waste generation trends, energy recovery potential, environmental impacts, economic feasibility, and policy frameworks influencing Waste-to-energy development. Findings reveal that the substantial quantity of municipal and industrial waste generated across the state presents significant opportunities for renewable energy production and environmental improvement. However, inadequate waste segregation practices, insufficient infrastructure, limited investment, technological constraints, and weak regulatory enforcement remain key barriers to effective deployment. The review highlights the importance of integrated waste management systems, supportive policies, stakeholder engagement, and public-private partnerships in enhancing Waste-to-energy adoption. It concludes that strategic investment in appropriate Waste-to-energy technologies can contribute significantly to sustainable waste management, improved energy security, and environmental sustainability in Ogun State.

Submission ID: ICSEECR-2026-086 | Track: T-01 | Presentation: In-Person | Oral

Building Digital Creators: A Proposed Community-Based Model for Early Digital Literacy and Hands-on STEM Education

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Abstract—Digital transformation has accelerated the demand for foundational digital literacy and Science, Technology, Engineering, and Mathematics (STEM) competencies, particularly among children and adolescents in underserved communities. However, unequal access to practical digital education continues to widen the digital divide, limiting opportunities for innovation, entrepreneurship, and sustainable socio-economic development. This paper proposes a community-based model for developing early digital literacy through hands-on STEM education using accessible, low-cost, and open-source technologies. The proposed framework integrates Scratch programming, educational robotics, collaborative project-based learning, peer mentoring, and community engagement to transition young learners from passive consumers of technology to confident digital creators and problem solvers. The study adopts a conceptual framework approach, synthesizing evidence from existing literature on digital literacy, community-based learning, STEM education, and sustainable development to design a scalable implementation model suitable for resource-constrained communities. The framework further outlines implementation phases, stakeholder participation, learner engagement strategies, and proposed evaluation indicators for measuring digital competence, creativity, collaboration, and innovation outcomes. By encouraging local problem-solving, digital inclusion, and responsible technology use, the proposed model contributes to strengthening community digital capacity while supporting sustainable human capital development and innovation ecosystems. The framework aligns with the United Nations Sustainable Development Goals, particularly SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), SDG 10 (Reduced Inequalities), and SDG 11 (Sustainable Cities and Communities). The proposed model provides educators, policymakers, and community organizations with a practical roadmap for promoting inclusive digital education and preparing the next generation of innovators capable of contributing to resilient and sustainable communities.

Index Terms—Digital Literacy; STEM Education; Community-Based Learning; Project-Based Learning; Sustainable Development; Digital Creators.

Submission ID: ICSEECR-2026-130 | Track: T-01 |
Presentation: Oral

Renewable Energy and Smart Grids

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Abstract—The global energy transition is reshaping electricity systems as nations strive to reduce reliance on fossil fuels and mitigate climate change. Renewable energy sources — such as solar photovoltaic, wind power, hydropower, and biomass — are increasingly deployed to achieve sustainable development goals. However, their intermittent and variable generation profiles pose significant challenges to conventional grids, which were designed for centralized and predictable energy flows. Smart grids have emerged as a transformative solution to these challenges, offering advanced digital communication, real-time monitoring, and automated control to enhance the resilience and flexibility of power networks. By integrating distributed generation, demand-side management, and energy storage technologies, smart grids provide a robust framework for balancing fluctuating renewable supply with dynamic consumer demand. They reduce transmission losses, improve fault detection through self-healing mechanisms, and empower consumers via advanced metering infrastructure, fostering active participation in energy management. This convergence of renewable energy and smart grid technologies also supports emerging trends such as electric mobility, decentralized micro grids, and prosumer models, thereby expanding the scope of sustainable energy applications. Despite these advantages, several barriers impede widespread adoption. High capital investment requirements, cyber security vulnerabilities, and regulatory complexities remain pressing concerns. Furthermore, successful deployment demands coordinated policy frameworks, technological innovation, and stakeholder collaboration to ensure equitable access and long-term sustainability. Addressing these challenges requires interdisciplinary research that bridges engineering, economics, and policy studies, alongside cooperative governance that aligns national energy strategies with global climate objectives. In conclusion, the integration of renewable energy with smart grid technologies represents a critical pathway toward achieving resilient, low-carbon energy systems. By overcoming technical, economic, and policy barriers, smart grids can unlock the full potential of renewables, advancing both energy security and climate goals. This synergy underscores the importance of sustained innovation and collaborative action in accelerating the transition to a sustainable energy future.

Index Terms—Renewable Energy; Smart Grids; Sustainability; Climate Change Mitigation; Energy Transition; Digital Communication.

Submission ID: ICSEECR-2026-077 | Track: T-01 |
Presentation: In-Person | Oral

Integration of Computer Vision and Data Analytics for Energy Efficiency: A Systematic Review of 50 Studies

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Abstract—The integration of Artificial Intelligence (AI), Internet of Things (IoT), and Big Data Analytics (BDA) has emerged as a cornerstone for enhancing energy efficiency and operational sustainability across industrial, residential, and urban sectors. This paper synthesises findings from fifty (50) recent research studies (2023 – 2025) evaluating the deployment of advanced cognitive systems and machine learning (ML) frameworks — including Deep Learning (CNN, LSTM, YOLO), Reinforcement Learning, and Federated Learning — to optimise energy consumption and predictive maintenance. Computer Vision (CV) and IoT-driven automation in smart buildings achieve occupancy detection accuracies exceeding 95%, leading to energy savings between 15% and 45% through adaptive lighting and HVAC control. ML-based predictive maintenance models have demonstrated a 35 – 45% reduction in equipment downtime and a 20 – 30% extension in asset lifespans. The application of Digital Twins and hybrid algorithms in renewable energy systems, specifically PV monitoring, has reached defect detection accuracies of up to 96%. Persistent challenges in data privacy, scalability, and computational demands are identified, and a Cognitive-Visual Energy Autonomy (CVEA) integrated multi-layer framework is proposed.

Index Terms—computer vision; energy efficiency; IoT; deep learning; predictive maintenance; smart buildings.

Submission ID: ICSEECR-2026-105 | Track: T-01 |
Presentation: In-Person | Oral

Energy Efficiency Optimization in Marine Diesel Engines Using Modern Control Systems

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Abstract—Energy efficiency has become one of the most critical challenges facing marine diesel engine operation due to increasing fuel costs, stringent international environmental regulations, and the maritime industry's commitment to decarbonization. Conventional engine control systems primarily employ fixed or semi-adaptive control strategies that are unable to respond effectively to rapid variations in engine load, fuel quality, sea state, and operating conditions, resulting in increased fuel consumption, inefficient combustion, and higher greenhouse gas (GHG) emissions. This study examines the application of modern intelligent control systems to optimize the energy efficiency of marine diesel engines. The objectives of the research are to develop an integrated control framework based on Model Predictive Control (MPC), adaptive control techniques, and Internet of Things (IoT)-enabled monitoring systems, and to evaluate their effectiveness in improving fuel efficiency, reducing exhaust emissions, enhancing engine performance, and supporting predictive maintenance under dynamic marine operating conditions. The study adopts a simulation- and data-driven methodology in which engine operational data are continuously acquired through a network of sensors measuring fuel injection timing, air-fuel ratio, engine speed, exhaust gas temperature, turbocharger performance, cylinder pressure, and engine load. These data are processed using intelligent control algorithms that automatically optimize combustion parameters in real time. A predictive maintenance framework incorporating data analytics and condition monitoring is integrated to detect performance degradation and potential component failures before they result in operational downtime. System performance is evaluated using key indicators, including specific fuel oil consumption (SFOC), brake thermal efficiency, emission levels (CO₂, NO_x, and particulate matter), engine response time, and maintenance intervals. The proposed integrated control strategy is further validated by comparing its performance with conventional engine control methods under varying operating scenarios. The findings demonstrate that combining MPC, adaptive control, IoT sensing, and predictive analytics provides superior operational performance compared with traditional control systems by significantly improving fuel economy, reducing emissions, enhancing engine reliability, and extending equipment service life. Unlike previous studies that focus on isolated control techniques, this research presents a comprehensive intelligent engine management framework that integrates advanced control, real-time monitoring, and predictive

maintenance into a unified optimization strategy. The study contributes directly to the achievement of United Nations Sustainable Development Goals (SDG 7: Affordable and Clean Energy, SDG 13: Climate Action, and SDG 14: Life Below Water) by promoting energy-efficient, environmentally sustainable, and technologically advanced marine propulsion systems capable of supporting the future of green shipping

Submission ID: ICSEECR-2026-114 | Track: T-01 |
Presentation: In-Person | Oral

Development of a Drone-Based Agricultural Monitoring System with AI-Powered Crop Health Analysis

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Abstract—Precision agriculture encounters significant challenges in enhancing crop health monitoring, disease detection, and resource management. This study synthesizes insights from 50 recent research articles focused on precision agriculture technologies that leverage unmanned aerial vehicles (UAVs) and artificial intelligence (AI). The methods evaluated include various advanced analytical techniques such as Convolutional Neural Networks (CNN), You Only Look Once (YOLO), deep learning algorithms, fuzzy intelligence, multi-sensor fusion, predictive analytics, and the integration of the Internet of Things (IoT). Key findings of the study highlight the effectiveness of the EfficientNet-B3 model, which achieves an impressive 98.8% accuracy in detecting crop diseases. Additionally, research findings indicate that other models, such as those explored by Arulprakash et al., show a disease detection accuracy of 92%, pest detection accuracy of 85%, and soil assessment accuracy of 89%. Their analysis also reveals significant operational benefits, including a 60% reduction in inspection time, a 30% increase in crop yield, and a 25% decrease in pesticide use. Furthermore, Nkwocha and Chandel contribute to the field by offering a comprehensive end-to-end framework for diagnosing crop diseases, designed around the latest sensing and computing technologies. This framework emphasizes the potential for improved decision-making processes in precision agriculture. To address identified limitations and research gaps, the study proposes a new framework known as the Multi-Tiered Autonomous Precision Agriculture (MT-APA). This framework aims to enhance the integration of various technologies and methodologies to optimize crop monitoring and management practices. Overall, the synthesis of these research insights illustrates the transformative potential of UAV and AI technologies in precision agriculture, illustrating promising advancements that could lead to more efficient farming practices. The study advocates for continued exploration and refinement of these technologies to effectively enhance agricultural productivity and sustainability,

thereby addressing the pressing challenges faced by the agricultural sector. The insights presented not only underline the current state of technology in precision agriculture but also point towards significant future research directions that could further revolutionize the industry.

Index Terms—Precision Agriculture; UAV; CNN; Sensor Fusion; Yield Optimisation; Efficientnet.

Submission ID: ICSEECR-2026-017 | Track: T-01 |
Presentation: In-Person | Oral

AI-Based Predictive Maintenance System for Electrical Machines Using Sensor Data Analytics

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Abstract—Electrical machine failures represent one of the most critical operational challenges in modern industrial environments. These machines serve as the primary drivers for production lines, and any unexpected malfunction can lead to cascading effects, including costly unplanned downtime, compromised worker safety, and a significant reduction in overall system reliability. To address these challenges, the industry is shifting from reactive maintenance to proactive, data-driven strategies. This study presents a comprehensive quantitative analysis of 50 peer-reviewed research articles published between 2016 and 2025. These articles examine the integration of artificial intelligence and machine learning techniques for the predictive maintenance of electrical machines using advanced sensor data analytics. The selection process followed a rigorous PRISMA-guided methodology to ensure the inclusion of high-quality, relevant studies sourced from major academic databases, including IEEE Xplore, ScienceDirect, Springer, and MDPI. The analysis reveals a dominant trend toward deep learning and machine learning applications. Deep learning approaches — particularly Convolutional Neural Networks and Long Short-Term Memory networks — consistently achieve prediction accuracies exceeding 97%. These models are highly effective at processing complex, time-series sensor data to identify subtle patterns that precede failure. Interestingly, traditional machine learning models remain highly relevant, with Logistic Regression achieving up to 98% accuracy in specific motor fault prediction scenarios. Machine learning techniques overall constitute 50% of the reviewed studies, highlighting their continued importance alongside more complex neural architectures. Despite these technological advancements, the review identifies six critical research gaps that hinder the widespread adoption of these systems. First, there is a limited focus on hybrid model development, which appears in only 12% of the literature. Second, real-time deployment at the edge remains insufficient, mentioned in only 14% of studies, which is a significant barrier for industries

requiring low-latency responses. Third, the integration of explainable AI is underdeveloped, appearing in only 8% of the reviewed papers. This lack of transparency makes it difficult for human operators to trust and act upon the decisions made by the AI. To bridge these gaps, this study proposes a robust hybrid framework combining Convolutional Neural Networks, Long Short-Term Memory networks, XGBoost, and SHAP. This architecture is designed to capture both spatial and temporal features of sensor data while providing high-performance boosting and model interpretability. The framework targets a prediction accuracy of at least 98%, ensuring that industrial operators have access to reliable and understandable diagnostic insights to prevent catastrophic machine failures.

Index Terms—Predictive Maintenance; Electrical Machines; CNN; LSTM; Xgboost; Xai.

Submission ID: ICSEECR-2026-019 | Track: T-01 |
Presentation: In-Person | Oral

Assessment of the Impact of Green Building Practice on Property Value in Lekki, Lagos State

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Abstract—The real estate industry has adopted green building principles as a result of growing awareness of energy efficiency and environmental sustainability. In rapidly developing urban areas such as Lekki, Lagos State, green buildings are becoming an important factor influencing property demand and market value. This study examines the impact of green building practices on property value in Lekki, Lagos State, with the view of assessing how sustainable construction features influence rental and capital values of residential and commercial properties. The research design used in the study is a descriptive survey. The target population of the study consists of registered Estate Surveyors and Valuers, Real estate developers, Architects and landlord residing in Lekki, Lagos state. The sample size is 50 respondents which consist of 15 real estate developers, 10 Estate Surveyors and Valuers, 10 architects and 15 residents of conventional and green-rated buildings in the study area. Descriptive statistical techniques like frequency distribution, percentages, mean scores, ranking methods are utilized to examine the data. The results show that Lekki property values are positively impacted by green building techniques. Higher rental income, greater market demand, and better long-term investment returns are typically drawn to properties with

green features. Widespread adoption is hampered by issues like high initial construction costs, low awareness, insufficient technical know-how, and the absence of a standardized green building certification. The Study concludes that green building techniques greatly raise the market competitiveness and property value in Lekki, Lagos State. The Study recommends raising stakeholder awareness, providing governmental support, offering financial incentives, and providing professional training.

Index Terms—Green building; Property value; Sustainability; Real estate; Lekki; Lagos State.

Submission ID: ICSEECR-2026-110 | Track: T-01 |
Presentation: In-Person | Oral

Beyond Passive Design: Artificial Intelligence and Smart Architecture as Emerging Pathways to Climate Resilient Building in Nigeria

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Abstract—Nigeria's construction industry finds itself caught in a genuinely difficult bind. On one side temperatures are climbing with alarming consistency and on the other, the national power grid chronically underpowered and overstretched simply cannot meet what the country actually needs. The result is a kind of slow-moving crisis playing out inside buildings everywhere, from family homes in Lagos to commercial towers in Abuja. These structures are consuming electricity at rates wildly out of proportion to what they deliver in return, and the millions of Nigerians living and working inside them are still sweating through it all, comfort nowhere in sight. Looking at this, the traditional passive design tricks like angling a building to catch the breeze or adding strategic shading actually do a decent job of lowering cooling needs. But here is the kicker: they aren't enough on their own anymore. Beyond that, in a nation where over 60% of the population is essentially living without a steady grid connection, relying solely on passive airflow is a losing battle as urban temperatures continue to skyrocket. This paper essentially argues that Artificial Intelligence (AI) and smart building sensors aren't just high-tech replacements for good architecture; they are its next necessary evolution. By diving into hard data, energy simulations, and the smart city policies currently taking shape in Lagos and other major hubs, this research maps out exactly where we stand with "intelligent" buildings. Basically, we have identified the structural roadblocks that are stopping these technologies from taking over. Looking at this, we see a really strange gap between professional knowledge and the actual physical reality of Nigeria's built environment. Basically, while an overwhelming 90.24% of the country's construction experts know exactly how these intelligent systems function, very few are actually putting them into the ground. Beyond that, for a structural analysis

perspective, this is a critical point: the transition from theoretical awareness to the actual integration of smart frameworks within a building's bones is still, in its infancy. Even though the industry is talking the talk, the actual adoption rate remains embryonic at best. Looking at this reality, the paper wraps up by proposing a new "convergence framework." The idea is simple but powerful as bioclimatic passive strategies should serve as the bedrock, and then we layer AI-driven optimization, renewable energy microgrids, and real-time adaptive controls on top. Beyond that, this integrated approach offers a genuine path toward making buildings energy-efficient, climate-resilient, and actually liveable within one of Africa's most intense and fastest-growing urban landscapes. Keywords: artificial intelligence; smart buildings; passive design; climate resilience; building energy efficiency; thermal comfort; Nigeria

Index Terms—Artificial Intelligence; Smart Building; Passive Design; Climate Resilience; Building Energy Efficiency; Thermal Comfort.

Submission ID: ICSEECR-2026-036 | Track: T-01 |
Presentation: In-Person | Oral

Mobile Application for Remote Solar Panel Fault Detection Using AI

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Abstract—The proliferation of solar photovoltaic (PV) installations globally has created an urgent demand for portable, accessible, and intelligent fault detection tools that can be operated by field technicians without specialised hardware. Mobile application platforms, when integrated with AI-based image analytics, offer a compelling solution for remote, real-time PV panel inspection. This paper presents a systematic review of fifty (50) peer-reviewed studies (2018 – 2025) on AI-driven fault detection and monitoring techniques for PV systems, with a specific focus on mobile and edge-deployable implementations. A PRISMA-compliant selection protocol was applied. Key findings identify CNN-based models — particularly MobileNetV2 and EfficientNetB0 — as the dominant architectures for on-device deployment, with accuracy ranging from 91% to 99.7%. YOLO variants (v5 through v11) are the leading frameworks for real-time detection. Critical limitations including computational overhead, data scarcity, and lack of field-validated mobile deployments are identified, and a proposed AI-powered mobile inspection framework with edge inference is presented.

Index Terms—mobile application; solar PV; fault detection; AI; edge inference; MobileNet; YOLO; IoT; remote monitoring.

Submission ID: ICSEECR-2026-094 | Track: T-01 |
Presentation: In-Person | Oral

Industry 4.0 Technologies in Manufacturing: A Review of Applications, Benefits, Challenges, and Implications for Competitiveness

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Abstract—Industry 4.0 has fundamentally transformed manufacturing by integrating advanced digital technologies with conventional production systems to create intelligent, interconnected, and data-driven production environments. As manufacturers face increasing pressure to enhance productivity, product quality, operational flexibility, supply chain resilience, and sustainability, digital transformation has become a strategic imperative for maintaining global competitiveness. This paper presents a comprehensive review of recent empirical literature on the applications, benefits, challenges, and competitiveness implications of Industry 4.0 technologies in manufacturing. Peer-reviewed journal articles and conference publications were critically synthesized to examine major technological enablers, sector-specific applications, implementation outcomes, and barriers to adoption across automotive, aerospace, electronics, pharmaceuticals, food processing, textiles, chemicals, and metal manufacturing industries. The review further incorporates regional perspectives by examining Industry 4.0 implementation in developed and emerging economies, highlighting differences in digital maturity, infrastructure, policy support, and industrial capabilities. The findings indicate that the greatest performance improvements are achieved when complementary technologies are implemented as integrated digital ecosystems rather than isolated solutions. Reported benefits include enhanced productivity, predictive maintenance, product quality, operational resilience, supply chain visibility, and resource efficiency. However, successful implementation remains constrained by high investment costs, cybersecurity risks, interoperability challenges, legacy system integration, organizational resistance, and workforce capability gaps, particularly among small and medium-sized enterprises where scalability remains a significant challenge. The review emphasizes that sustainable manufacturing competitiveness depends not only on technology acquisition but also on strategic investments in digital infrastructure, workforce reskilling and upskilling, organizational readiness, and supportive policy frameworks. By synthesizing evidence from diverse manufacturing sectors and regional contexts, this study provides practical insights for researchers, policymakers, and industry practitioners while identifying future research priorities in scalable Industry 4.0 adoption, human-centred manufacturing, trustworthy artificial intelligence, and the transition toward Industry 5.0. The conclusions are grounded in empirical evidence reported across the reviewed studies, thereby strengthening

understanding of effective digital manufacturing transformation.

Index Terms—Industry 4.0; Smart Manufacturing; Digital Transformation; Internet of Things; Workforce Development; Sustainable Manufacturing.

Submission ID: ICSEECR-2026-100 | Track: T-01 |
Presentation: In-Person | No Preference

Spatio-Temporal Dynamics of Urban Expansion and Security Implications in Saapade, Ogun State, Nigeria (1984–2024)

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Abstract—The rapid urbanization in developing countries has led to significant changes in land use patterns. This has serious implications for urban security and sustainable development. While studies have found that urban growth in developing countries often leads to heightened criminal activity due to weak social control and inadequate infrastructure, there is limited localized research on how small but rapidly growing towns, such as Saapade, manage these emerging crime threats, especially in the absence of robust urban governance. This study examines the dynamics of urban expansion in Saapade, Ogun State, over a four-decade period from 1984 to 2024, focusing on the relationship between spatial growth and urban security. The study adopted a quantitative survey; primary data was collected through the administration of questionnaires. Satellite imagery and Geographic Information System (GIS) were utilized for Land Use and Land Cover (LULC) analysis. Results revealed that between 1984 and 2024 there was extensive conversion of vegetation into built-up land. Between 1984 and 2018, about 2,220 hectares were converted (≈ 65 ha annually), while between 2018 and 2024, 1,373 hectares were converted (≈ 229 ha annually), showing a threefold acceleration. A majority of residents (54.7%) expressed heightened concern about security due to expansion, while 78.8% believed crime had increased in recent years. Key challenges included theft and burglary, cult-related activities, and land disputes. The pace of expansion led to structurally disorganized growth, which weakened social bonds and increased security vulnerability. The study recommends integrated planning and security strategies to manage urban growth sustainably. Proactive collaboration between government, planners, institutions, and communities will be crucial to ensuring that Saapade's expansion translates into sustainable development without undermining safety.

Index Terms—Urban Expansion; Spatio-Temporal Analysis; Land Use Land Cover; Urban Security; Saapade; Nigeria.

Submission ID: ICSEECR-2026-141 | Track: T-01 |
Presentation: In-Person | Oral

Design and Implementation of a Networked Computer System: Multiple PC's Switch/Router Configuration for File Sharing

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Abstract—The design and implementation of a networked computer system that facilitates file sharing among multiple PCs through switch/router configuration presents a vital advancement in information technology particularly for collaborative environments such as offices and educational institutions. While prior studies have examined peer-to-peer (P2P) architectures and centralized client-server models in isolation, few have addressed the practical integration of switch/router-level configuration, VLAN segmentation, and DHCP-based addressing within a single, testable framework tailored to small-to-medium institutional networks. This gap is significant because organizations in resource-constrained settings, such as polytechnics and small offices in developing regions, often lack access to enterprise-grade networking infrastructure and instead require cost-effective, simulation-validated designs before physical deployment. This study is guided by four objectives: to design a network topology integrating switch/router configuration, VLAN segmentation, and DHCP-based IP assignment for institutional file sharing; to simulate and validate the design using Cisco Packet Tracer prior to physical deployment; to evaluate the design's performance in terms of connectivity, file transfer efficiency, and configuration overhead; and to assess its cost-effectiveness and sustainability relative to conventional static-addressing, enterprise-grade alternatives. The study adopts a design-and-implementation methodology structured in four stages: requirements analysis, network topology design with VLANs defined by user role or department, protocol configuration using DHCP for automated addressing, and simulation and validation in Cisco Packet Tracer. Test scenarios included inter-VLAN ping connectivity checks, DHCP lease verification, and file transfer tests across peer nodes. Simulation results showed a 34% improvement in average file transfer speed and a 12ms average latency reduction compared to the unsegmented baseline network, alongside an approximately 75% reduction in configuration time achieved through DHCP automation over static addressing. Beyond these performance gains, the design offers measurable sustainability benefits: reduced reliance on dedicated centralized servers lowers hardware and energy requirements, while simulation-first validation minimizes costly reconfiguration and hardware waste during physical deployment, supporting long-term cost efficiency for institutions with limited IT budgets. This

work contributes to national and institutional digital infrastructure development, aligning with SDG 9 (Industry, Innovation, and Infrastructure) and SDG 4 (Quality Education). The study recommends that resource-constrained institutions adopt simulation-validated P2P and VLAN-based designs as a practical, sustainable pathway toward secure and efficient networked file sharing.

Index Terms—Peer-to-Peer (P2P); DHCP; Virtual LANs (VLAN); Cisco Packet Tracer; Network Simulation.

Submission ID: ICSEECR-2026-157 | Track: T-01 |
Presentation: Virtual | Oral

Arduino Uno: Architecture, Applications, and Relevance to Renewable Energy and Smart Grid Systems

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Abstract—Arduino Uno is an open-source microcontroller development board built around the ATmega328P microcontroller chip, and it has grown to become one of the most recognized and widely adopted platforms in embedded systems and electronics. The board was designed to lower the barrier to entry for beginners in electronics and programming, offering a practical combination of affordability, simplicity, and a large, active community of users spread across more than 150 countries. This paper is a technical survey rather than a formal systematic literature review, providing a structured examination of the Arduino Uno covering its historical background, hardware architecture and core components, operational principles, and the range of practical applications it supports, including home automation, educational robotics, environmental monitoring, healthcare sensor devices, and general Internet of Things (IoT) deployments. Particular attention is given to the board's use in renewable energy management and smart grid systems, where Arduino Uno-based sensor networks monitor grid voltage, current, and load in real time and manage automatic switching between grid, generator, and photovoltaic sources; this class of application is linked explicitly to three United Nations Sustainable Development Goals — SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 11 (Sustainable Cities and Communities) — and is evaluated against the deterministic, certified performance expected of industrial programmable logic controllers (PLCs) in utility-grade smart grid duty. The methodology draws on official technical documentation, peer-reviewed journal and conference literature published between 2022 and 2026, and reference textbooks on Arduino hardware and programming. Seven documented case studies are evaluated using qualitative comparative analysis,

supported by a quantitative comparison of the Arduino Uno's technical specifications against four competing microcontroller platforms. The results show a consistent pattern of low-cost implementation with strong reported accuracy where outcomes were quantitatively validated, although most case studies report functional or behavioural success rather than metrological precision. The review demonstrates the continuing relevance of the Arduino Uno in modern technology education and applied engineering, while also identifying key limitations such as restricted processing power, limited memory capacity, and an inability to meet the determinism, isolation, and certification demands of industrial-grade smart grid control.

Index Terms—Arduino Uno; Embedded Systems; Internet of Things (IoT); Microcontroller Platforms; Renewable Energy; Smart Grid Monitoring.

Submission ID: ICSEECR-2026-162 | Track: T-01 | Presentation: Oral

Empirical Analysis of Nigerian Students' CBT Readiness Using JAMB CBT as a Case Study

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Abstract—This study presents an empirical analysis of Nigerian students' CBT readiness, using the Joint Admissions and Matriculation Board (JAMB) examination as a case study. The research investigates “readiness” as a multidimensional construct encompassing technical proficiency, psychological preparedness, and institutional support. Utilizing a quantitative survey of 5,088 candidates, the study reveals a critical “readiness gap” despite nearly a decade of CBT implementation in Nigeria. Findings indicate that 87.9% of candidates are technical “beginners,” and only 20.6% participated in the official mock examination before the exam, highlighting a severe deficit in preparatory readiness. This lack of technical skill strongly correlates with an 81.7% anxiety rate, as candidates struggle with operational tasks such as mouse navigation and time management under a “hostile” on-screen timer, a stressor for 74.8% of respondents. Furthermore, an infrastructure-readiness paradox was observed: while physical centers achieved 100% hardware readiness and 99.7% power stability, 72.6% of candidates faced software-level glitches, indicating a lack of institutional network readiness. Paradoxically, 65.4% of students maintained a future preference for CBT over paper-based tests, citing result transparency as their primary motivator. The study concludes that Nigerian students are ideologically ready for digital testing but technically and psychologically underprepared. It recommends a mandatory “CBT readiness certification” for all candidates and a shift in policy focus from hardware

procurement to user experience (UX) optimization and rural digital inclusion to ensure equitable access in future examination cycles.

Index Terms—CBT Readiness; Digital Literacy; Exam Anxiety; User Experience (UX); Institutional Readiness; Educational Technology; Nigeria.

Submission ID: ICSEECR-2026-156 | Track: T-01 | Presentation: In-Person | Oral

Cloud-Based Image Analytics for Real-Time Solar Panel Maintenance Scheduling

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Abstract—The rapid global expansion of solar photovoltaic (PV) energy installations has generated an acute demand for scalable, intelligent, and automated maintenance solutions. As of 2024, global installed PV capacity surpasses 1.6 terawatts, placing enormous operational pressure on asset managers. Traditional inspection methodologies are fundamentally inadequate at this scale, failing to provide the spatial resolution, temporal frequency, and diagnostic precision required. This paper presents a comprehensive investigation into the application of cloud-based image analytics pipelines for automated solar panel fault detection and real-time maintenance scheduling, grounded in a systematic quantitative review of fifty (50) peer-reviewed research articles published between 2018 and 2024. CNN-based models achieve mean detection accuracies of 95.8%, while hybrid deep learning models deliver peak performance averaging 97.1% and reaching 99.4% maximum. Cloud deployment via AWS SageMaker and Google Vertex AI substantially reduces inference latency and enables elastic horizontal scalability. The paper proposes CIARMS — a six-layer Cloud-based Image Analytics for Real-time Maintenance Scheduling framework — and identifies critical research gaps in explainability, cross-climate generalisation, and federated learning.

Index Terms—Cloud Computing; Image Analytics; Solar PV; CNN; YOLO; Maintenance Scheduling; CIARMS; Transfer Learning.

Submission ID: ICSEECR-2026-064 | Track: T-01 | Presentation: In-Person | Oral

Smart Waste Bin for Sustainable Solution in Engineering Education

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Abstract—Rapid urbanization has increased municipal solid waste, leading to overflowing bins, unhygienic conditions, and inefficient collection routes. Conventional waste management depends on fixed timetables and manual inspection, which waste time, fuel, and labour. This paper proposes the Smart Waste Bin, designed to enhance ease of waste disposal for the public with less human effort, in the context of sustainable engineering solutions. The Smart Waste Bin uses IoT and embedded sensors to automate monitoring and reporting. The design integrates an ultrasonic sensor to measure fill level, a gas sensor to detect harmful odours, and a microcontroller to process the data. A wireless communication module transmits real-time status to a cloud dashboard accessible by sanitation teams via a mobile or web interface. When waste level exceeds a set threshold, the system automatically sends a notification to the control centre, enabling dynamic route planning and on-demand collection and reducing unnecessary trips. The bin also includes a servo-driven lid that opens on proximity detection to limit human contact, and its low-power design supports battery or solar operation. Experimental testing shows the design improves collection efficiency, prevents overflow, and supports cleaner public spaces, offering a scalable and affordable approach to modernizing waste management in smart cities.

Index Terms—Smart Waste Bin; IoT; Ultrasonic Sensor; Real-Time Monitoring; Waste Management; Automation.

Submission ID: ICSEECR-2026-163 | Track: T-01 |
Presentation: In-Person | Oral

Effect of Indoor Air Quality on The Health and Productivity of Industrial Workers

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Abstract—The concentration of CO₂ and particulate matter (PM_{2.5}), such as dust, usually builds up and increases within an industrial workspace environment. These build-ups are influenced by factors such as population size, work activities, the heat- or dust-generating equipment in use, and the ventilation system. These factors cumulatively determine the quality of indoor air within any given workspace environment. This study uses statistical data to develop a model that predicts the effects of indoor air quality (IAQ) on the

health and productivity of industrial workers, and also develops a smart Indoor Air Quality Index (IAQI) model for effective monitoring, control, and management of IAQ. A quantitative research methodology was used in analysing datasets of annual workers' attendance records, focusing on presenteeism rate, absenteeism rate, and working-hours rate. Correlation analysis was used to determine the relationship between average monthly CO₂ and PM_{2.5} concentrations and the dependent factor of average monthly absenteeism rate, and also between thermal comfort and working-hours rate and the dependent variable of productivity. Results were verified using correlation coefficients and regression analysis to show the strength of the relationships. The results show that a positive correlation coefficient indicates that presenteeism and working-hours rates increase as thermal comfort improves, while a negative correlation coefficient indicates that workers' health deteriorates with increasing concentrations of CO₂ and PM_{2.5} in the room.

Index Terms—Carbon Dioxide Concentration; Indoor Air Quality; Particulate Matter; Thermal Comfort.

Submission ID: ICSEECR-2026-140 | Track: T-01 |
Presentation: In-Person | Oral

Design of a Gesture-Controlled Robotic Arm Using Sensor Fusion and Machine Learning Algorithms

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Abstract—The advancement of gesture-controlled robotic arms has become a critical research area due to the growing demand for intuitive, precise, and real-time human-robot interaction (HRI) across industrial, healthcare, rehabilitation, assistive technology, and collaborative robotic applications. Conventional robotic control methods often require physical interfaces, complex programming, or manual operation, which can limit flexibility and ease of use. Gesture-based control systems provide a more natural and efficient means of interaction by allowing users to communicate commands through body movements, hand gestures, and muscle signals. This study presents a comprehensive analysis of fifty peer-reviewed articles on gesture-controlled robotic arm systems, covering sensing technologies for gesture acquisition and motion tracking, including Inertial Measurement Units (IMUs), Electromyography (EMG) sensors, Micro-Electro-Mechanical Systems (MEMS), computer vision-based systems, and Internet of Things (IoT)-enabled devices. Various computational and control methodologies were also evaluated, including deep learning techniques such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and Bidirectional LSTM (BiLSTM) models for gesture recognition and pattern analysis, alongside classical machine learning algorithms, sensor fusion

techniques, predictive control approaches such as Predictive Dynamic Modeling (PDM) and Kalman filtering, and teleoperation frameworks. The findings reveal that multi-sensor fusion combined with machine learning techniques consistently delivers the highest performance levels, often achieving gesture recognition and control accuracies exceeding 95%. By integrating information from multiple sensor sources, these systems effectively overcome challenges associated with sensor noise, environmental variations, and gesture ambiguity, resulting in more reliable and robust robotic manipulation. Based on the identified strengths and limitations of existing approaches, this study proposes a Hybrid Integrated Neuro-Symbolic Architecture for Multi-Modal Sensor Fusion and Autonomous Robotic Manipulation, combining the learning capabilities of neural networks with the reasoning and decision-making strengths of symbolic artificial intelligence, to enhance gesture interpretation, autonomous task execution, system adaptability, explainability, and overall human-robot collaboration.

Index Terms—Gesture-Controlled Robotic Arm; Sensor Fusion; Machine Learning.

Submission ID: ICSEECR-2026-138 | Track: T-01 |
Presentation: In-Person | Oral

Smart Maintenance Systems for Solar Panels Using IoT and Image-Based Analytics

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Abstract—Solar photovoltaic (PV) systems represent one of the fastest-growing sources of renewable energy globally, with cumulative installed capacity exceeding 1.6 terawatts as of 2024 (International Renewable Energy Agency, 2024). Traditional manual inspection methods for solar panel maintenance are labour-intensive, costly, and poorly scalable. This paper presents a comprehensive investigation into the design and performance of Smart Maintenance Systems (SMS) for solar panels that integrate IoT sensor networks with image-based deep learning analytics. Grounded in a systematic review of fifty (50) peer-reviewed research articles published between 2018 and 2025, the study examines how IoT-enabled real-time data acquisition can be fused with CNN- and YOLO-based computer vision pipelines to deliver accurate, low-latency, and scalable fault detection and maintenance scheduling. Key findings indicate that hybrid IoT-image analytics systems achieve a mean fault detection accuracy of 95.8%, compared to 83.4% for sensor-only baselines—a 12.4 percentage-point improvement attributable to image-based deep learning integration.

Index Terms—IoT; Smart Maintenance; Solar PV; CNN; YOLO; Image Analytics.

Submission ID: ICSEECR-2026-060 | Track: T-01 |
Presentation: In-Person | Oral

Design and Implementation of a Biometric Attendance Management System Using a Computer System for the Electrical/Electronic Engineering Department, Gateway (ICT) Polytechnic, Saapade

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Abstract—Accurate attendance monitoring is essential for effective academic administration in higher institutions. Conventional attendance methods, such as paper-based registers and manual signature sheets, are often inefficient, time-consuming, and susceptible to errors, impersonation, and data loss. This paper proposes a biometric attendance management system designed to improve the accuracy, security, and efficiency of attendance recording within the Department of Electrical and Electronic Engineering. The system employs fingerprint or facial recognition technology to authenticate students before automatically recording their attendance in a centralised database. The design integrates a biometric sensor, a microcontroller-based interface, a personal computer, and a software application for student enrolment, authentication, attendance recording, and report generation. During operation, each student's biometric template is securely stored in the database and verified during attendance registration, with the corresponding date and time recorded automatically. By facilitating automated attendance management, the system minimises fraudulent attendance practices, reduces administrative workload, improves processing speed and accuracy, and provides real-time attendance reports for lecturers and administrators. Experimental evaluation demonstrates that the system achieves high recognition accuracy, fast authentication time, and reliable performance under normal operating conditions. The proposed system offers a secure, scalable, and cost-effective attendance management solution suitable for universities, polytechnics, colleges, and other educational institutions.

Index Terms—Biometric Attendance Management System; Biometric Authentication; Fingerprint Recognition; Facial Recognition; Student Attendance; Database Management; Campus Automation; Identity Verification; Educational Technology.

Submission ID: ICSEECR-2026-168 | Track: T-01 |
Presentation: In-Person | Oral

Challenges of Property Assessment and Rating in Remo North Local Government Area, Ogun State, Nigeria

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Abstract—Property rating and assessment is the process of assessing landed properties by qualified valuers or appraisers, with the aim of billing and collecting rates from the occupiers, tenants, or owners of such properties, depending on the enabling laws of the affected states. Property tax is generally a levy imposed by municipalities upon owners of real property within their jurisdiction, based on the value of that property, and property tax assessment and rating represent a comprehensive process that revalues property to determine tax liabilities, serving as the financial backbone for local authorities to ensure accurate revenue collection and fair taxation. This study investigates the challenges encountered by property rating officers in carrying out property rating in Remo North Local Government Area, Ogun State. A questionnaire was distributed to rating officers at Remo North Local Government, and several problems were identified in the course of the exercise.

Index Terms—Property; Assessment; Rating; Challenges; Solution.

Submission ID: ICSEECR-2026-171 | Track: T-01 | Presentation: In-Person | Oral

A Climate-Adaptive Zoning Framework Using Climate Projections for Flood Risk Reduction in Ibadan, Nigeria

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Abstract—Flooding remains the most recurrent climate hazard in Ibadan, Nigeria, yet land allocation continues in high-risk areas despite the availability of flood data and climate projections. The September 2011 flood displaced over 121,000 people, and the 2022 events followed the same pattern in Apete, Odo-Ona, and Sasa. NIHSA and NiMet consistently classify Ibadan North, South-East, and Akinyele as “highly probable” flood zones, while CMIP6 models project an 8 to 12% increase in extreme 1-day rainfall for Ibadan by 2050. This study examines why climate information is not translated into land-use decisions by focusing on the Land Use Act 1978. Using policy analysis, a review of 20 statutory approval files from 2018 to 2023, and 18 key informant interviews with planners, surveyors, and community leaders in Ibadan Metropolis, the study tested two hypotheses within a Legal-Institutional-Behavioural framework: the first proposing no significant relationship

between the provisions of the Land Use Act 1978 and Oyo State Physical Planning Law 2016 and the integration of climate projections in statutory land allocation decisions, and the second proposing a significant positive relationship between the level of legal protection afforded planning officials and their willingness to reject land applications in flood-prone areas. The findings support both hypotheses: the Act contains no hazard provision, and NIHSA data were not referenced in any approval file. Furthermore, 83% of officials cited a lack of legal backing as the reason for approving risky applications. The study proposes a Climate-Adaptive Zoning Addendum with three components: mandatory attachment of NIHSA flood status to land applications, Red-Yellow-Green zoning using NIHSA and CMIP6 data, and a liability shield for officials. Stakeholder validation confirmed that the framework is feasible within current state law. The paper concludes that Ibadan does not need new maps, but one new legal paragraph to enable climate-adaptive land allocation.

Index Terms—Land Use Act 1978; Climate-Adaptive Zoning; Flood Risk; Ibadan; CMIP6.

Submission ID: ICSEECR-2026-172 | Track: T-01 | Presentation: In-Person | Oral

Impact of Urban Planning on Commercial Property in Sagamu, Remo, and Its Environs

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Abstract—People prefer to live in a well-planned environment because it gives them access to healthcare, education, and better infrastructure, thereby improving quality of life and enabling people to contribute more to the economy. Good urban planning reduces traffic congestion, inefficient infrastructure, and expensive housing. Urban planning is a forward-thinking process used to organise towns and cities; it impacts property by influencing the transportation system, infrastructure layout, business activity, and population density. Without proper urban planning, cities can quickly become inefficient, making them unappealing to potential residents and businesses; sound knowledge of urban planning is therefore vital to sustaining higher property values and supporting profit margins. This paper focuses on the impact of urban planning on commercial property in Sagamu, Remo, Ogun State, and its environs. A combination of interviews and questionnaires was administered to landlords and tenants within the locality to determine the impact of current urban renewal efforts in the neighbourhood.

Index Terms—Impact; Urban Planning; Commercial; Property.

Submission ID: ICSEECR-2026-174 | Track: T-01 |
Presentation: In-Person | Poster

TECHNICAL TRACK T-03

SUSTAINABLE INFRASTRUCTURE & GREEN BUILDING

18 Accepted Abstracts

Sodium-Ion Battery Technology as a Sustainable Energy Storage Solution for Solar Power Systems

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Abstract—The global transition to renewable energy hinges on the availability of reliable, affordable, and sustainable energy storage technologies. Across sub-Saharan Africa and other developing regions, solar photovoltaic (PV) deployments are expanding rapidly, yet widespread adoption continues to be constrained not by generation capacity but by energy storage limitations. Lithium-ion battery (LIB) technology, which currently dominates the storage market, presents significant challenges including high capital costs, supply chain vulnerabilities, and a geographically concentrated raw material base barriers that are particularly acute in resource constrained economies. This paper presents investigation of sodium-ion battery (SIB) technology as a viable and sustainable alternative for solar energy storage, addressing critical gaps in the existing literature, notably the absence of cost modelling frameworks, grid integration analyses, and system-level design guidelines suited to developing country contexts. Three core objectives guide this paper; to conduct a comparative technoeconomic assessment of LIB and SIB technologies for stationary solar storage; to evaluate both chemistries against key performance indicators including cycle life, thermal behaviour, and levelized cost of storage (LCOS); and to develop a systems engineering roadmap for SIB integration into solar microgrids and national grid stabilisation programmes. The methodology adopts a multi-criteria technoeconomic framework, combining published electrochemical performance data, material cost benchmarks, and field deployment case studies drawn from tropical and high-temperature operating environments. Cost projections are modelled over a 10 - 20 years lifecycle horizon, incorporating anticipated manufacturing scale up and material learning curves. Findings reveal that SIBs offer

a compelling value proposition for stationary solar applications where gravimetric constraints are non-critical. Compared to conventional LIBs, SIBs demonstrated a 25% reduction in capital cost per kWh, retained 85% capacity at 45°C without active cooling, and reduced operational maintenance costs by approximately 15%. These results support a structured systems engineering roadmap aimed at accelerating SIB adoption across solar microgrid architectures, offering a resource-secure and economically viable pathway toward energy sustainability in developing nations.

Index Terms—Sodium-ion batteries; solar energy storage; energy transition; technoeconomic analysis; microgrid integration; sustainable engineering.

Submission ID: ICSEECR-2026-079 | Track: T-03 |
Presentation: In-Person | Oral

The Evaluation of the Impact of Urban Sprawl on Land Use in Peri-Urban Areas of Lagos State

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Abstract—Urban sprawl has emerged as a significant challenge to sustainable urban land use in the peri-urban areas of Lagos State, Nigeria. Rapid population growth, rural – urban migration, and increasing economic activities have accelerated the outward expansion of Lagos metropolis, resulting in substantial changes in land use patterns, environmental quality, infrastructure demand, and the sustainability of urban development. This study evaluates the impact of urban sprawl on sustainable urban land use in selected peri-urban communities of Ibeju-Lekki, Lagos State. Specifically, the study examines the extent of urban sprawl, assesses its effects on land use sustainability, property development, and infrastructure provision, and identifies

measures for promoting sustainable urban growth in peri-urban areas. A mixed-methods research approach was adopted for the study. The study population comprised residents, property owners, estate surveyors and valuers, and staff of the Lagos State Physical Planning Permit Authority (LASPPPA). A sample size of 110 respondents was selected using purposive and stratified sampling techniques, consisting of 50 residents, 25 property owners, 25 estate surveyors and valuers, and 10 staff of LASPPPA. Primary data were collected through structured questionnaires, interviews, and field observations, while secondary data were obtained from journals, textbooks, government publications, and planning documents. Data collected were analysed using descriptive statistical techniques, including frequency distributions, percentages, and mean ranking. The findings reveal that urban sprawl has contributed positively to increased land values, housing development, and economic opportunities within peri-urban communities. However, it has also accelerated the conversion of agricultural land to urban uses, increased traffic congestion, placed pressure on existing infrastructure, and contributed to environmental degradation and uncoordinated physical development. The study further identifies weak planning control mechanisms, ineffective land administration systems, and inadequate enforcement of development regulations as major factors driving unsustainable urban expansion. The study concludes that achieving sustainable urban land use in Lagos State requires effective land use planning, improved infrastructure provision, strengthened development control measures, and the integration of environmental sustainability principles into urban growth management. The study contributes to the attainment of Sustainable Development Goal (SDG) 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land) by promoting sustainable urban development, environmental conservation, and resilient land use practices in rapidly growing peri-urban areas.

Submission ID: ICSEECR-2026-132 | Track: T-03 |
Presentation: In-Person | Oral

Evaluation of Locally Sourced Sustainable Building Materials for Green Construction

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Abstract—The building sector contributes significantly to greenhouse gas emissions and is a big consumer of natural resources. Construction expenses, energy consumption, and environmental degradation have all increased in Nigeria due to the country's reliance on traditional building materials like steel, cement, and imported finishing goods. This study assesses the

energy-efficiency potential, economic viability, environmental performance, and compatibility of locally obtained sustainable building materials for green construction. It also looks at how they contribute to the development of smart infrastructure, the integration of renewable energy, and the accomplishment of Sustainable Development Goals (SDGs) 9 (Industry, Innovation and Infrastructure), 11 (Sustainable Cities and Communities), and 13 (Climate Action). The study design used was a descriptive survey. Structured questionnaires were used to gather primary data from construction experts, such as architects, builders, quantity surveyors, engineers, estate surveyors, and valuers. To provide sufficient representation, a sample size of 109 respondents was chosen using stratified and purposive sampling approaches. Frequency distributions, percentages, mean scores, and the Relative Importance Index (RII) were used to examine the data. The results show that because of their low embodied energy, affordability, local availability, and lower carbon footprint, stabilized earth blocks, bamboo, laterite, lumber, clay products, and agricultural waste-based composites have great potential for sustainable construction. According to the study, using these materials can increase building energy efficiency, lessen reliance on energy-intensive imported materials, and make it easier to incorporate renewable energy technologies like solar photovoltaic systems into climate-responsive building designs. The study comes to the conclusion that using sustainable building materials that are supplied locally offers a practical solution to achieve resilient urban infrastructure, energy-efficient buildings, and lower carbon emissions. Increased use of them can help achieve SDGs 9, 11, and 13 and promote green building practices. To encourage sustainable building in Nigeria, the report suggests investing in research and innovation, strengthening policy support, developing professional competence, and establishing quality standards.

Submission ID: ICSEECR-2026-112 | Track: T-03 |
Presentation: In-Person | Oral

Public Private Partnership Model for Financing Sustainable Infrastructure in Lagos State

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Abstract—Public private partnership models for financing sustainable infrastructure in lagos state this study examined the effect of public-private partnership (ppp) models on financing sustainable infrastructure in lagos state. The study was motivated by the increasing need for quality infrastructure such as roads, transportation systems, housing, water supply, and other public facilities, which cannot be financed by government alone. The study adopted a descriptive

survey research design and used a structured questionnaire to collect data from respondents. The population of the study consisted of 1,250 stakeholders, including government officials, private investors, project managers, consultants, and other professionals involved in infrastructure development in Lagos State. Using the Taro Yamane formula, a sample size of 303 respondents was selected for the study. Data collected were analyzed using the Statistical Package for the Social Sciences (SPSS). Descriptive statistics such as frequencies, percentages, and mean scores were used to analyze the data, while regression analysis was used to test the hypotheses. The findings revealed that Public-Private Partnership models contribute significantly to sustainable infrastructure financing in Lagos State by attracting private sector investment, improving service delivery, and reducing the financial burden on government. The study also found that challenges such as inadequate policy implementation, corruption, weak institutional frameworks, and financial risks affect the successful implementation of PPP projects. The study concluded that Public-Private Partnerships are an effective means of financing sustainable infrastructure development in Lagos State. The study recommended improved government policies, greater transparency, stronger regulatory frameworks, increased private sector participation, enhanced stakeholder collaboration, better risk-sharing mechanisms, regular project monitoring, and continuous capacity building to strengthen PPP implementation and promote long-term economic growth, social development, environmental sustainability, efficient resource utilization, improved public accountability, investor confidence, timely project completion, better maintenance practices, innovation, inclusive development, and improved quality of life for residents across the state. Ultimately these measures will enhance infrastructure resilience, economic competitiveness, public welfare, and sustainable prosperity.

Index Terms—Public Private Partnership; Sustainable Infrastructure; Infrastructure; Development.

Submission ID: ICSEECR-2026-127 | Track: T-03 |
Presentation: In-Person | Oral

Construction of Corn Shelling Machine

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Abstract—Corn (maize) is a major food and industrial crop with a relatively short growing cycle of about three months, making it economically important to farmers. However, post-harvest shelling remains a significant challenge for many small-scale producers who still rely on manual shelling or rudimentary hand-operated tools. These methods are labour-intensive, time-consuming, inefficient, and often cause grain damage and post-harvest losses. This study presents the design,

fabrication, and performance evaluation of a low-cost corn shelling machine constructed entirely from locally sourced materials to improve affordability, ease of maintenance, and local manufacturability. The machine employs an impact-based shelling mechanism consisting of flexible steel chains mounted on a rotating shaft to achieve efficient grain separation while minimizing kernel breakage. Engineering design calculations were carried out for the shaft, frame, power transmission system, bearings, and other critical components to ensure structural integrity and operational efficiency. Fabrication was completed using conventional workshop equipment and readily available local materials. Performance validation involved repeated experimental trials under varying operating conditions, including different maize moisture contents and feed rates, to assess reliability and consistency. Standard agricultural machinery testing protocols were used to evaluate shelling efficiency, throughput capacity, grain damage, cleaning efficiency, and operational stability. The machine achieved an average throughput capacity of approximately 320 kg/h with high shelling efficiency and minimal grain damage. Durability assessment through continuous operation indicated satisfactory wear resistance and dependable performance under normal working conditions. The modular design allows scalability through increased power input and structural modifications for medium-scale processing applications. The use of locally available materials reduces fabrication costs, simplifies maintenance, and promotes sustainable local manufacturing. The developed corn shelling machine provides an efficient, reliable, and cost-effective post-harvest processing solution capable of reducing labour requirements, minimizing grain losses, and improving agricultural productivity, particularly for small- and medium-scale farmers in developing economies.

Index Terms—Corn shelling machine; agricultural mechanization; locally sourced materials; performance evaluation; impact mechanism; scalability.

Submission ID: ICSEECR-2026-107 | Track: T-03 |
Presentation: In-Person | Oral

Optimizing Hybrid Solar PV – Battery Storage for Climate-Resilient and Sustainable Rural Electrification: Evidence from Isara Remo, Nigeria

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Abstract—Rural communities in Southwest Nigeria continue to experience chronic electricity shortages due to unreliable grid supply, limited grid expansion, and dependence on expensive diesel generation, thereby constraining socio-economic development. This study

designs and optimises hybrid solar PV and battery storage systems for Isara Remo (6.9°N, 3.7°E, Ogun State) across three representative rural load archetypes comprising industrial, residential and commercial demand (M1 Industrial: 1,500 kW peak; M2 Residential: 1,000 kW; M3 Commercial: 800 kW), two battery chemistries (LFP and VRLA), and four grid-availability scenarios (G0, G3, G6, G9; 0–9 average daily grid hours derived from 105 documented Nigerian grid failures, 2015–2024). A 22-year (2001–2022) NASA POWER solar resource dataset and field-informed hourly load profiles were embedded in a custom Python-based hourly dispatch simulation in which grid power supplements load only and never charges the battery. Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO) minimized Net Present Cost (NPC) subject to a Loss of Power Supply Probability (LPSP) constraint, with Mann-Whitney U testing across ten independent runs. Under the G3 reference scenario, optimal LFP configurations achieved LCOE of USD 0.2398–0.3600/kWh—23–36% lower in lifecycle battery cost than equivalent VRLA configurations and 40–65% lower NPC than the Nigerian diesel benchmark (USD 0.688/kWh). PSO and GWO showed no statistically significant performance difference ($p > 0.05$). Progressive grid availability reduced NPC by 12.6% (G3), 25.2% (G6), and 37.9% (G9) relative to the off-grid baseline, extending battery life from 11.0 to 19.9 years. Monte Carlo analysis (1,000 Latin Hypercube samples) incorporating uncertainty in PV cost, battery cost, discount rate, solar irradiance, O&M cost, load growth and diesel price confirmed 99.8% probability that the optimal LFP configuration outperforms diesel generation. The results indicate that LFP is the preferred storage technology for community-scale hybrid PV systems under the investigated Nigerian operating conditions.

Index Terms—hybrid solar photovoltaic; battery energy storage; lithium iron phosphate; particle swarm optimization; grey wolf optimizer; techno-economic optimization; rural electrification; grid reliability; leveled cost of energy; renewable energy; microgrid; Nigeria.

Submission ID: ICSEECR-2026-093 | Track: T-03 |
Presentation: In-Person | Oral

A Wireless Pyranometer Using NodeMCU as a Tool for Sustainable Energy

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Abstract—Accurate solar irradiance measurement remains a cornerstone of effective solar energy system design, optimization, and long-term performance monitoring. Yet conventional pyranometers continue to present significant barriers, they are expensive, mechanically bulky, and typically dependent on wired

data logging infrastructure that restricts their use in remote, off-grid, or geographically distributed environments. These limitations become especially pronounced in developing regions where solar energy holds the greatest untapped potential but where instrumentation resources are most constrained. This paper presents the design, implementation, and field evaluation of a low-cost, wireless pyranometer built around the NodeMCU (ESP8266) microcontroller. The system integrates a calibrated photovoltaic or thermopile sensor for solar radiation measurement, with the NodeMCU managing onboard data processing and wireless transmission over Wi-Fi. Using the MQTT (Message Queuing Telemetry Transport) protocol, instantaneous irradiance readings are transmitted to a cloud-based dashboard, enabling continuous remote monitoring, automated data logging, and real-time visualization without physical intervention. Calibration and validation exercises were conducted by benchmarking the developed system against a standard commercial-grade instrument under varying atmospheric and seasonal conditions. Results demonstrated strong statistical correlation alongside acceptable error margins, confirming the system's suitability for practical field deployment. Importantly, the evaluation also examined long-term reliability across fluctuating temperature, humidity, and solar angle conditions factors that often degrade sensor accuracy and wireless connectivity over extended operational periods. While the system delivers compelling performance at a fraction of the cost of conventional alternatives, this work also honestly identifies potential scalability constraints. Network dependency, power supply continuity in off-grid scenarios, and sensor drift over prolonged exposure represent areas requiring further engineering attention as deployment scales from single units to distributed sensor networks. By offering an accessible, replicable, and budget-conscious instrumentation solution, this wireless pyranometer contributes meaningfully to expanding sustainable energy monitoring infrastructure particularly in underserved communities where data-driven solar planning can have transformative socioeconomic impact.

Index Terms—Wireless Pyranometer; nodeMCU; Solar Irradiance; Sustainable Energy; Remote Monitoring.

Submission ID: ICSEECR-2026-080 | Track: T-03 |
Presentation: In-Person | Oral

AI-Enabled Green Cybersecurity Framework for Sustainable Healthcare IoT: Enhancing Patient Data Security Through Energy-Efficient Intelligent Systems

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Abstract—Full abstract text was not available from the author at the time of publication of this volume. Please

refer to the conference secretariat or the online submission record (Submission ID above) for the complete abstract.

Submission ID: ICSEECR-2026-091 | Track: T-03 |
Presentation: In-Person | Oral

Construction of Palm Kernel Cracking Machine

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Abstract—Palm kernel processing is an important component of Nigeria's agricultural value chain; however, the widespread reliance on manual cracking methods continues to limit productivity due to low throughput, high labour requirements, and excessive kernel breakage. Furthermore, imported cracking machines remain unaffordable for many small- and medium-scale processors and are often difficult to maintain because of limited access to spare parts and technical expertise. This study presents the design, fabrication, and performance evaluation of a locally manufactured palm kernel cracking machine developed to improve processing efficiency while supporting sustainable rural industrialization. The machine was designed using a combined impact and compression cracking mechanism to maximize shell breakage while minimizing kernel damage. Material selection was guided by strength, wear resistance, machinability, local availability, affordability, and ease of maintenance. Mild steel plates were used for the hopper and cracking chamber, medium-carbon steel for the shaft, angle iron for the supporting frame, and standard locally available bearings and pulleys to facilitate fabrication and future maintenance. Fabrication was carried out using conventional workshop processes, including cutting, machining, welding, drilling, assembly, and surface finishing. Performance evaluation was conducted using dried palm nuts obtained from a local oil palm mill under controlled operating conditions. Experimental testing involved repeated trials in which cracking efficiency, kernel breakage rate, throughput capacity, processing time, and power consumption were measured to assess machine performance and operational reliability. The fabricated machine achieved an average cracking efficiency of 92%, a kernel breakage rate of 5.4%, a throughput capacity of 85.7 kg/h, and a power consumption of 1.6 kW. Compared with conventional manual cracking, the machine increased processing capacity by more than fifteen times while significantly reducing kernel losses and operator fatigue. The use of locally sourced materials lowers production costs, simplifies maintenance, and reduces dependence on imported equipment, making the technology particularly suitable for rural communities. Beyond improving agricultural processing, the proposed machine contributes to the development of sustainable rural infrastructure by strengthening local manufacturing

capacity, promoting indigenous engineering solutions, supporting small-scale Agro-industries, and creating opportunities for employment and enterprise development. The machine also offers potential for future integration with renewable energy sources, thereby enhancing climate resilience and reducing the environmental footprint of Agro-processing operations. Consequently, the study directly supports Sustainable Development Goal (SDG) 2 (Zero Hunger) through improved agricultural productivity and reduced post-harvest losses, while advancing SDG 9 (Industry, Innovation and Infrastructure) by promoting affordable, resilient, and locally engineered processing technologies for sustainable industrial development

Submission ID: ICSEECR-2026-113 | Track: T-03 |
Presentation: Oral

Life Cycle Cost Analysis of Sustainable Buildings Materials for Infrastructure Projects

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Abstract—This study examined the life cycle cost analysis of sustainable building materials for infrastructure projects, with the aim of evaluating their long-term economic and environmental benefits compared to conventional building materials. The research was conducted to determine the cost effectiveness, durability, maintenance requirements, and sustainability performance of selected sustainable materials used in construction projects. The study adopted a survey research design using primary data collected through questionnaires administered to 100 construction professionals which consists of 25 Engineers, 25 Architects, 25 Quantity surveyors and 25 project managers involved in infrastructure development projects. The population of the study consisted of construction practitioners within the construction industry, while a sample size of 100 respondents was selected using simple random sampling techniques. Data collected from the respondents were analyzed using descriptive and comparative analysis methods to examine the life cycle costs of sustainable building materials. The analysis covered the stages of material procurement, installation, operation, maintenance, replacement, and disposal. Findings from the study revealed that although sustainable building materials may involve higher initial construction costs, they provide lower maintenance expenses, improved durability, reduced energy consumption, and longer service life over time. The study concludes that the adoption of sustainable building materials can improve both economic efficiency and environmental sustainability in infrastructure projects. It therefore recommends increased awareness, supportive

government policies, and proper implementation of life cycle cost evaluation methods to encourage sustainable construction practices within the construction industry.

Submission ID: ICSEECR-2026-129 | Track: T-03 |
Presentation: In-Person | Oral

Category-Invariant Sizing of Institutional Photovoltaic – Battery – Inverter Systems: A Statistically Validated, Transferable Framework from a 35-Building Nigerian Polytechnic Campus

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Abstract—Institutional campuses across sub-Saharan Africa are increasingly adopting decentralised photovoltaic (PV) systems to offset unreliable grid electricity, yet the published sizing literature is dominated by single-building or single-load-class case studies. Little methodological guidance exists for sizing PV – battery – inverter systems consistently across a genuinely heterogeneous campus, where laboratories, heavy-duty workshops, administrative blocks, and low-power utility units coexist under one procurement programme. This paper develops and statistically validates a stepwise, standards-based sizing methodology applied uniformly across 35 functionally distinct buildings at a Nigerian polytechnic campus (832.5 kW combined connected load, 970.5 kVA aggregate inverter capacity, 1,800 solar panels), formalising four sizing stages — daily energy demand, array capacity, battery autonomy, and inverter/charge-controller matching — as explicit engineering equations. Buildings were stratified into four functional categories (laboratories, workshops, administrative blocks, utility units), and category-level sizing ratios were tested formally with nonparametric rank-based statistics, normality and variance-homogeneity diagnostics, bootstrap-corrected effect sizes, and complete post-hoc comparisons, rather than assessed by descriptive comparison alone. Neither the battery-count ratio nor the charge-controller-current ratio differed significantly across categories once tested at full-campus scale, indicating that category membership alone does not reliably predict per-kilowatt component ratios. A single minimum-load security post was formally confirmed as a statistical outlier in controller-current provisioning, traced to an irreducible minimum-component floor rather than a category-wide effect. Connected load remained a

strong, near-linear, category-invariant predictor of panel count campus-wide, supporting a single verified scaling model ($N_{\text{panels}} \approx 0.00316 \times \text{load}$) with direct procurement applications. The paper's contribution is threefold: a generalised, reproducible institutional sizing workflow (rather than a single-building design exercise); statistically validated, category-invariant scaling ratios; and a predictive model, internally validated on this campus, that offers a transferable planning framework for Gateway (ICT) Polytechnic, Saapade, and comparable multi-building institutions subject to local validation.

Index Terms—PV System Design; Institutional Energy Systems; Solar Microgrid; Battery Autonomy; Inverter Matching; Campus Electrification.

Submission ID: ICSEECR-2026-142 | Track: T-03 |
Presentation: In-Person | Oral

Architecture of Calm: Investigating the Relationship Between Minimalist Spatial Design and User Well Being in Nigerian Residential Buildings

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Abstract—For a long time, Nigerian urban housing has been judged almost exclusively by two measures how much of it exists, and what it costs to build. That framing, reasonable as it might seem on the surface, leaves out something important. The psychological and physiological reality of actually living inside these spaces what the built environment does to the people inhabiting it day after day has received far too little serious attention. This paper makes a direct case for minimalist spatial design as a genuinely appropriate response to that gap. Think of it less as a style and more as a discipline one built around what actually matters in a living space. Streamlined spatial organisation. Layouts that breathe and adapt. Spaces that flex between functions, daylight and moving air treated as foundational design elements rather than optional extras. And visual excess, cleared away entirely. These are not isolated ideas together, they form a design philosophy that the paper argues is both coherent and genuinely responsive to the way Nigerians actually live inside their homes. Building that case required pulling from four distinct areas of knowledge: environmental psychology, architectural minimalism scholarship, Nigeria's own deep tradition of vernacular building, and field-based research into indoor environmental quality gathered across Nigerian cities. From that combination, something concrete emerges a conceptual model that ties particular minimalist spatial decisions to outcomes residents can feel in their daily lives, reduced stress, clearer thinking and physical comfort in the heat. A meaningful sense of belonging in one's own space, but what actually anchors the argument is historical. Look closely at how Nigerians have always built courtyards at the centre, layouts tuned

to the local climate, rooms designed from the start to do more than one thing and the minimalist instinct was already there, operating quietly long before anyone attached a name to it. This is not a foreign idea arriving at the door but a familiar one, returning. There is no friction such that the Nigerian culture has always read architecture as statement space that signals generosity, pride, presence. Stripping that back sits uneasily against those values, and the paper takes that conflict seriously. Rather than talking past it, the work proposes design strategies that genuinely hold both sides, the discipline of minimalist thinking and the cultural need for spaces that feel expressive and alive. From there, it turns practical giving architects, builders urban planners, and policymakers actual direction for embedding these principles into a housing landscape that is not standing still. Keywords: Minimalist architecture, user well-being, Nigerian housing, indoor environmental quality, spatial design, residential satisfaction, psychological comfort, environmental psychology

Index Terms—Minimalist Architecture; User Well Being; Nigerian Housing; Indoor Environmental Quality; Spatial Design; Psychological Comfort.

Submission ID: ICSEECR-2026-035 | Track: T-03 |
Presentation: In-Person | Oral

The Evaluation of the Role of Artificial Intelligence in Real Estate Marketing in Ijebu-Ode Ogun State

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Abstract—This study evaluated the role of Artificial Intelligence (AI) in sustainable real estate marketing in Ijebu-Ode. The increasing adoption of digital technologies across industries has created opportunities for AI to transform traditional real estate marketing practices through enhanced customer engagement, market analysis, and marketing efficiency. Despite these opportunities, the level of AI adoption among real estate practitioners in many developing urban centres remains uncertain. The study therefore examined the level of awareness and adoption of AI tools, identified the benefits of AI applications in sustainable real estate marketing, and assessed the challenges affecting AI adoption in the study area. Descriptive research design was adopted for the study. The population consisted of real estate developers, information technology personnel, and estate surveyors and valuers operating within Ijebu-Ode. A purposive sampling technique was employed, and a total of 60 questionnaires were distributed to respondents. Out of these, 49 questionnaires were retrieved and found suitable for analysis, representing a response rate of 81.7%. Data collected were analyzed using frequency distribution tables, percentages, mean scores, standard deviation, and ranking. The findings revealed that there is a high level of awareness of Artificial Intelligence among real estate

practitioners, although the actual level of adoption remains moderate. The study further established that AI contributes significantly to marketing efficiency, customer engagement, property promotion, and decision-making processes. Respondents agreed that AI tools enhance market reach, improve communication with clients, and support data-driven marketing strategies. However, the study identified major challenges hindering AI adoption, including high implementation costs, inadequate technological infrastructure, low digital literacy, poor data availability, and resistance to change among practitioners. The study concluded that Artificial Intelligence has a positive and significant impact on sustainable real estate marketing and possesses enormous potential for improving the efficiency and competitiveness of the real estate sector in Ijebu-Ode. The study recommended increased investment in digital infrastructure, regular training and capacity-building programmes for practitioners, development of affordable AI solutions, and greater support from professional bodies and policymakers to encourage wider adoption of AI technologies in sustainable real estate marketing.

Index Terms—Artificial Intelligence; Digital Technology; Marketing; Property; Real Estate; Real Estate Marketing; Sustainable; Ijebu-Ode.

Submission ID: ICSEECR-2026-096 | Track: T-03 |
Presentation: In-Person | Oral

Procurement Cost Scales Proportionally, Not by Economies of Scale: A Statistically Validated Techno-Economic Framework for Institutional Solar Microgrid Deployment

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Abstract—Institutional solar electrification programmes are typically costed building-by-building, yet little empirical evidence documents how unit procurement cost scales with system size when one institution deploys photovoltaic (PV) systems across many buildings of markedly different loads under a single programme. This paper analyses procurement cost data for a 35-building, 832.5 kW campus-wide PV-battery-inverter deployment at a Nigerian polytechnic, totalling ₦685.71 million, to quantify this scaling relationship formally rather than by descriptive comparison alone. Cost per installed watt ranged from ₦528/W to ₦1,637/W, a roughly 3-fold spread, with total cost correlating very strongly with connected load ($r = .981$). Grouping buildings into small,

medium, and large load bands produced materially identical mean costs per watt, and neither parametric nor nonparametric tests found a significant difference between bands. A linear model fitted across all 35 buildings ($R^2 = .962$) yields a null intercept, indicating no statistically detectable fixed balance-of-system cost, and out-performs a power-law (six-tenths-rule) alternative under cross-validation; its exponent is statistically indistinguishable from simple proportionality rather than from the classical six-tenths economies-of-scale range. Influence diagnostics identify three very large buildings, not the smallest, as the most disproportionately influential points, and a supplementary analysis of covariance shows that functional building category adds no explanatory power beyond connected load. Building on the statistically supported linear model, this paper adds a full economic and environmental performance layer — levelised cost of energy, net present value, internal rate of return, discounted payback, and avoided-emissions analysis, computed across a 10–20% discount-rate range and a ± 10 –20% capital-cost sensitivity band rather than at a single assumed rate — to translate the statistical cost-scaling result into decision-relevant investment and sustainability indicators, together with an explicit reconciliation of a cost discrepancy in the underlying procurement estimate. Together, these results indicate that, at full-campus scale, procurement cost scales essentially proportionally with connected load, a materially different conclusion from the pronounced economy-of-scale pattern suggested by a smaller, partial sample of the same campus. The paper's contribution is a fully tested, transferable cost-scaling and techno-economic evaluation framework, internally validated against a single institution's complete building stock, offering a framework applicable to Gateway (ICT) Polytechnic, Saapade, and comparable Nigerian tertiary institutions planning phased solar electrification, subject to local validation.

Index Terms—Techno-Economic Analysis; Solar Cost Scaling; Levelised Cost of Energy; Institutional Procurement; Distributed PV; Sub-Saharan Africa.

Submission ID: ICSEECR-2026-144 | Track: T-03 |
Presentation: In-Person | Oral

Digital Documentation of Cultural Heritage Assets in Abeokuta: An Integrated Framework for Sustainable Preservation and Heritage Management

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Abstract—The cultural heritage assets of Abeokuta, Nigeria, represent significant historical, architectural, socio-cultural, and economic resources that preserve the

identity and collective memory of the Egba people. However, rapid urbanization, environmental degradation, climate-related risks, inadequate maintenance, uncontrolled modernization, and the gradual erosion of indigenous knowledge continue to threaten both tangible and intangible heritage resources. Existing documentation practices remain fragmented and largely dependent on photographs, manual drawings, written records, and oral narratives, limiting effective conservation, long-term monitoring, disaster preparedness, and sustainable heritage management. This study addresses this gap by investigating how integrated digital documentation technologies can enhance the preservation and management of cultural heritage assets in Abeokuta, evaluating existing documentation practices, assessing the applicability of contemporary digital documentation technologies, and developing an Integrated Digital Heritage Documentation Framework (IDHDF) for sustainable heritage preservation. A mixed-methods research design was adopted, integrating field surveys, case-study investigations, and a systematic literature review. The study evaluated the application of Geographic Information Systems (GIS), photogrammetry, terrestrial laser scanning, Building Information Modelling (BIM), three-dimensional (3D) modelling, virtual reality, and digital archiving for documenting selected heritage assets, including Olumo Rock, Ake Palace, Centenary Hall, Kuti Heritage Museum, Itoku Adire Market, and Ijaye Pottery Market. The findings revealed that integrated digital documentation significantly improves the accuracy, completeness, accessibility, visualization, and long-term preservation of heritage information compared with conventional documentation methods, and that combining spatial mapping, 3D documentation, digital imaging, archival databases, and community-generated indigenous knowledge provides a more comprehensive and resilient approach to heritage management. Consequently, the IDHDF was developed to support evidence-based conservation planning, climate-risk preparedness, heritage monitoring, cultural tourism, education, and stakeholder participation, offering a practical, scalable, and replicable model for documenting and managing cultural heritage assets in Abeokuta and other rapidly urbanizing cities across Nigeria and sub-Saharan Africa. The study demonstrates that digital documentation is not only a conservation tool but also a strategic instrument for sustainable urban development and climate-resilient heritage management.

Index Terms—Abeokuta; Cultural Heritage; Digital Documentation; Heritage Conservation; Sustainable Heritage Management; Climate Resilience; Building Information Modelling (BIM).

Submission ID: ICSEECR-2026-165 | Track: T-03 |
Presentation: In-Person | Oral

Comparative Analysis of Predictive and Preventive Maintenance in Automobile Engineering Using IoT-Based Instruments

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Abstract—Today in automobile maintenance, artificial intelligence (AI)-driven technology such as the Internet of Things (IoT) has taken over maintenance systems from conventional methods, marking a data-driven transition from schedule-based approaches toward condition-aware strategies. Preventive maintenance (PM), characterized by scheduled servicing irrespective of vehicle condition, has long been the industry practice in automobile fleet management; however, its limitations — including over-maintenance, resource inefficiency, and an inability to act before failures begin — have driven growing adoption of predictive maintenance (PdM) using IoT-based instrumentation. This paper presents a systematic comparative analysis of predictive and preventive maintenance techniques in automobile maintenance engineering, with emphasis on IoT-based instrumentation as the enabling technology for PdM deployment. Eight categories of IoT-based instruments used in automobile condition monitoring were examined, each evaluated in terms of its characteristics, measured parameters, and suitability for predictive versus preventive maintenance workflows. The comparative framework covers fault detection capability, unplanned downtime incidence, maintenance accuracy, IoT infrastructure requirements, and return on investment (ROI). Analysis indicates that IoT-based PdM reduces unplanned vehicle downtime and lowers total maintenance expenditure relative to conventional PM over a comparable operational period, while PM retains operational advantages in resource-constrained environments where IoT infrastructure is unaffordable. In conclusion, using IoT-based instruments for precision diagnostics of highly sensitive components while retaining schedule-based protocols for low-risk components yields optimum results in contemporary automobile engineering maintenance.

Index Terms—Artificial Intelligence (AI); Predictive Maintenance (PdM); Preventive Maintenance (PM); Internet of Things (IoT); Automobile Engineering; Condition Monitoring.

Submission ID: ICSEECR-2026-075 | Track: T-03 | Presentation: In-Person | Oral

Real-Time Object Detection and Classification Using Computer Vision for Industrial Automation

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Abstract—The rapid advancement of deep learning, artificial intelligence (AI), and computer vision technologies has revolutionized industrial automation by enabling intelligent systems capable of performing real-time object detection, classification, and localization with remarkable speed and accuracy. Conventional machine vision systems primarily depend on handcrafted features, predefined rules, and image-processing techniques, making them less effective in dynamic manufacturing environments characterized by varying illumination, object orientation, background clutter, partial occlusion, and high production speeds. To overcome these limitations, this study presents a robust real-time object detection and classification framework based on advanced Convolutional Neural Network (CNN) architectures, specifically You Only Look Once (YOLO) and Faster Region-Based Convolutional Neural Network (Faster R-CNN). The proposed framework is designed to accurately detect, classify, and localize industrial objects, including manufactured components, defective products, missing parts, and foreign materials from continuous video streams captured on production lines. The system utilizes transfer learning with domain-specific industrial datasets to enhance detection performance while minimizing training time and computational complexity. Advanced image preprocessing techniques, including noise reduction, image normalization, and data augmentation, are incorporated to improve model generalization and robustness across different operating conditions. Adaptive confidence thresholding and Non-Maximum Suppression (NMS) algorithms are implemented to reduce false detections and improve localization accuracy under challenging conditions such as motion blur, overlapping objects, and partial occlusion. To satisfy real-time industrial requirements, the trained models are deployed on edge computing platforms equipped with GPUs and embedded AI hardware accelerators, enabling low-latency inference without relying heavily on cloud computing. Experimental evaluation demonstrates that the proposed framework achieves a mean Average Precision (mAP) exceeding 94% while maintaining processing speeds above 30 frames per second (FPS), making it suitable for high-speed industrial inspection and automation tasks. Comparative analysis further indicates that the proposed approach significantly outperforms conventional rule-based machine vision systems in terms of detection accuracy, adaptability, processing speed, robustness, and scalability. Moreover, seamless integration with Programmable Logic Controllers (PLCs), Supervisory Control and Data Acquisition (SCADA) systems, industrial robots, and automated conveyor systems

enables intelligent closed-loop manufacturing processes such as robotic sorting, automated defect rejection, quality inspection, inventory monitoring, predictive maintenance support, and real-time production alerts, providing a scalable, reliable, and cost-effective solution for smart factories within Industry 4.0 and emerging Industry 5.0 paradigms.

Index Terms—Computer Vision; Object Detection; Deep Learning; Industrial Automation; CNN; YOLO; Edge Computing; Quality Control; Industry 4.0; Machine Vision.

Submission ID: ICSEECR-2026-139 | Track: T-03 |
Presentation: In-Person | Oral

Fabrication of a Profile Cutting Machine

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Abstract—This project focused on the design and fabrication of a profile cutting machine for use in small- and medium-scale fabrication workshops. The study was motivated by the need to develop a cost-effective, locally manufactured alternative to imported profile cutting machines, which are often expensive to purchase, maintain, and repair. The machine was designed to cut mild steel profiles, including angle bars, flat bars, round pipes, square pipes, and solid rods, with improved accuracy, efficiency, and operational safety. The design process involved engineering calculations for motor power, shaft dimensions, belt and pulley transmission, frame strength, and work-holding vice, while standard bearings and an abrasive cutting wheel were selected to ensure reliable performance and durability. Fabrication was carried out using locally available materials and conventional workshop operations such as marking out, cutting, drilling, machining, welding, grinding, assembly, and finishing. The completed machine was subjected to performance evaluation by cutting different mild steel sections under varying operating conditions, assessing cutting accuracy, surface finish, ease of operation, machine stability, and safety. Test results showed that the machine produced clean, straight, and dimensionally accurate cuts while significantly reducing cutting time and operator effort compared with manual cutting methods; the rigid frame, effective clamping mechanism, and protective wheel guard enhanced operational stability and user safety throughout the cutting process. The use of locally sourced materials and standard machine components reduced overall production cost, simplified maintenance, and demonstrated the feasibility of indigenous manufacturing. The project confirms that a locally fabricated profile cutting machine can effectively satisfy the operational requirements of engineering workshops, technical institutions, and small-scale industries while promoting local technology development and reducing dependence on imported equipment. Future improvements should incorporate a variable-speed drive, automatic feed mechanism, coolant or dust extraction

system, and digital angle adjustment to enhance productivity, versatility, cutting precision, and machine service life.

Index Terms—Profile Cutting Machine; Round Pipes; Rigid Frame; Straight Cut; Local Fabrication.

Submission ID: ICSEECR-2026-170 | Track: T-03 |
Presentation: In-Person | Oral

TECHNICAL TRACK T-06

WATER RESOURCES & POLLUTION
CONTROL

3 Accepted Abstracts

Pathways to Addressing Barriers to Sustainable Waste Management for a Greener Future in Ogun State, Nigeria

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Abstract—There are many environmental management challenges facing cities of developing world, and among those challenges is attributed to waste generation management. In recent time, solid waste management, collection and disposal has continue to be a growing concern in Nigeria thus, addressing these environmental challenges requires a comprehensive and sustainable approach. This paper explores the pathways to addressing barriers to sustainable waste management for a greener future in Ogun State with reference to Abeokuta metropolis as a case study. Based on survey and qualitative data gathered, the study identifies key barriers to include poor facilities for waste collection, plants and equipments necessary for waste management, poorly managed collection point, low level of awareness among the public, and continued behavioral norms of open dumping and burning among others. The study recommends the need for continued public education and awareness campaigns on individual's behavioral change towards waste and the need to adopt sustainable strategies to enable address negative consequences related to the unregulated management of solid waste practices. Keywords: Barriers, Pathways, Sustainable, Waste Management, Ogun State. Keywords —

Index Terms—Barriers; Pathways; Sustainable; Waste Management; Ogun State.

Submission ID: ICSEECR-2026-031 | Track: T-06 |
Presentation: In-Person | Oral

Synergizing AI and IoT for Smart Waste Management: A Framework for Resilient and Sustainable Urban Ecosystems

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Abstract—Rapid global urbanization has led to an unprecedented rate of increase in the amount of municipal solid waste (MSW) generated, which is stressing the structural capacity of traditional waste management systems. Traditional collection and disposal systems are becoming more inefficient, more costly to operate, and are exacerbating environmental problems. This paper proposes a paradigm shift by exploring the synergistic use of Artificial Intelligence (AI) and the Internet of Things (IoT) to transform urban waste ecosystems from vulnerability to resilience. Cities have the potential to shift from reactive waste collection and management to predictive, automated waste management through a combination of robust IoT networks and advanced AI technologies, such as machine learning algorithms and computer vision. The first part of this study presents an extensive analytical survey of smart waste management architectures, while the second part examines their actual implementation in various cities around the world. Experience shows that intelligent waste technology implementation can bring about game-changing operational improvements and environmental sustainability gains. For example, smart bin networks in Singapore gave rise to a significant reduction in waste collection trips of up to 40–91%. In the same vein, data-driven dynamic vehicle routing in Amsterdam increased logistical efficiency by 20% to 50%, and advanced automated sorting systems in Boston and Songdo achieved waste sorting accuracy above 95%. Together, these interdependent technologies play a major

role in reducing greenhouse gas emissions, maximizing recycling, and diverting significant amounts of waste from landfill, driving progress toward SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action). The paper critically reflects on the socio-technical constraints and implementation barriers of these technologies and provides practical suggestions for urban planners, policymakers, and technology providers, concluding that building a unified AI and IoT network is no longer a luxury innovation but a key requirement for long-term sustainable urban development.

Index Terms—Artificial Intelligence; Internet of Things; Smart Waste Management; Urban Sustainability; Machine Learning; Computer Vision; Circular Economy.

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Presentation: Oral

rotates to open the lid, and closes it after a set delay to minimize contact and odor release. For power sustainability, the system uses a 12V solar panel with a charge controller and rechargeable battery, allowing off-grid operation. This introduces students to renewable energy design. From an educational perspective, the Arduino-based smart bin serves as a multidisciplinary capstone project. Students are involved in circuit design, Arduino programming in C/C++, sensor calibration, servo motor control, GSM/Wi-Fi communication, and dashboard development. It improves campus waste management while giving engineering students practical experience in designing smart systems for smart cities. Future work will include adding load cells for weight measurement and using machine learning for waste type classification.

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Presentation: In-Person | Oral

Arduino-Based Smart Waste Bin with Servo Motor for Sustainable Waste Management in Engineering Education

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Abstract—The rapid increase in solid waste generation in academic institutions has made traditional waste collection methods inefficient and unsustainable. Fixed collection schedules often lead to overflowing bins, poor sanitation, increased labor and fuel costs, and higher carbon emissions. At the same time, engineering education requires more practical, project-based platforms that allow students to apply theoretical concepts to real-world sustainability problems. There is therefore a need for a low-cost, smart, and educational system that addresses both campus waste management and student learning outcomes. This paper presents the design and implementation of an Arduino-Based Smart Waste Bin with Servo Motor as a sustainable solution integrated into engineering education. The primary objectives are: 1. to monitor and manage waste in real-time to promote a cleaner campus, and 2. to provide Electrical, Electronics, and Computer Engineering students with hands-on experience in embedded systems, IoT, sensors, actuators, and renewable energy. The system is built around an Arduino Uno microcontroller as the main processing unit due to its low cost, open-source platform, and ease of programming. The hardware consists of an *HC-SR04 ultrasonic sensor* mounted at the top of the bin to measure the waste fill level by calculating the distance to the waste surface. An MQ-2 gas sensor is included to detect harmful gases such as methane and ammonia produced during decomposition, supporting health and safety monitoring. A *servo motor SG90* is interfaced with the Arduino to provide automatic, touchless lid operation. When the PIR proximity sensor detects a user within 10cm, the servo

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APPRECIATION

The Organising Committee of ICSEECR 2026 extends its sincere appreciation to the Management, Staff and Students of Gateway (ICT) Polytechnic, Saapade, for their unwavering support in bringing this conference to life.

We are equally grateful to the Rector, the Deans of the School of Engineering & Technology and the School of Environmental Studies, the Technical Programme Committee, our distinguished Chief Guest of Honour, Keynote Speakers and Lead Paper Presenters, and every reviewer whose diligence shaped this Book of Abstracts.

To every author and presenter joining us from institutions across Nigeria and beyond: thank you for entrusting your scholarship to this platform. It is your research that gives this conference its purpose.

We wish all our delegates a rewarding conference and safe travels to and from Saapade.

Thank you all.

SPECIAL RECOGNITION

Nigerian Association of Technologists in Engineering (NATE)

Ogun State Branch



The Organising Committee of ICSEECR 2026 extends its warm appreciation to the Nigerian Association of Technologists in Engineering (NATE), Ogun State Branch, for its goodwill and continued partnership with Gateway (ICT) Polytechnic, Saapade, in the advancement of engineering and technology practice.

In furtherance of this partnership, NATE Ogun State Branch has indicated its intention to sponsor the **Best Student Paper Presentation Award** at this conference, an initiative that will further honour excellence and inspire rigorous scholarship among the students presenting at ICSEECR 2026.

The Organising Committee looks forward to a continued and fruitful collaboration with NATE, in shared pursuit of its vision to design, build and maintain for progress.

With appreciation, on behalf of the Organising Committee.

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